



Swinburne University of Technology Hawthorn Campus Department of Computing Technologies

COS20028 Big Data Architecture and Application Assignment 2 - Semester 2, 2024

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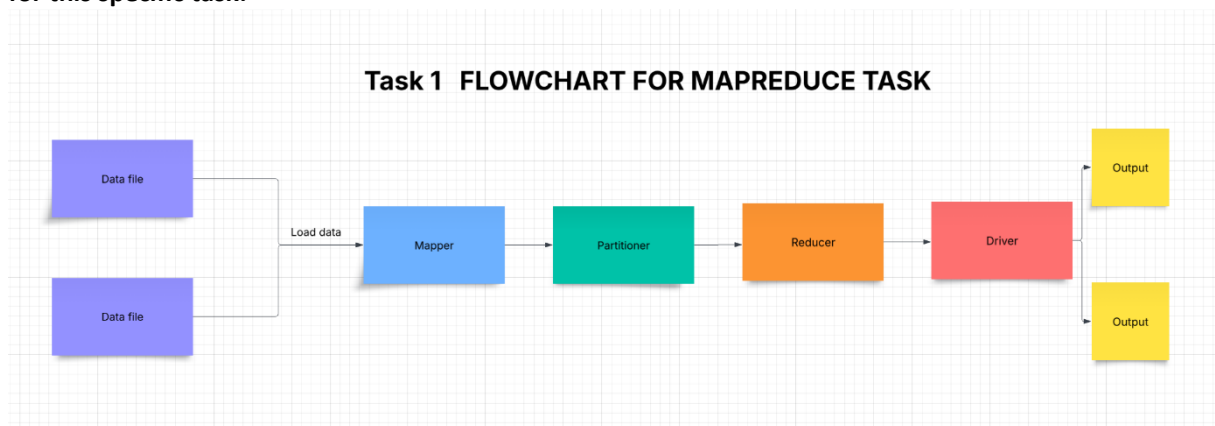
Submission Date: [20/7/2025](#)

Assignment Task

• Part 1

1. Explain how you manage the tool you chose to access multiple files in one go.

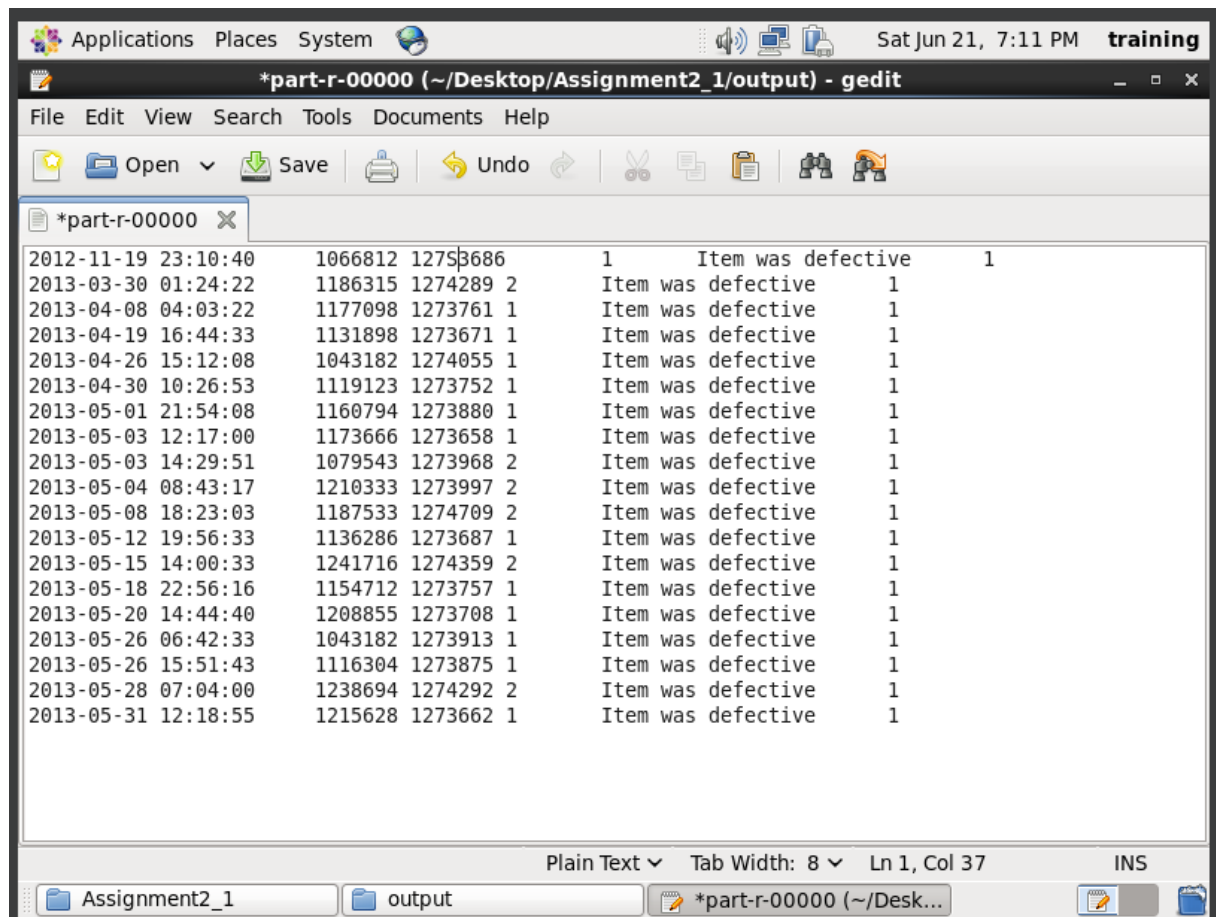
First, I decided that Hadoop MapReduce going to be the main tool that I'm going to use. Next step I'm planning the draft of Hadoop mapreduce workflow based on lesson learned on Week 6. It's Mapper => Partitioner => Reducer => Driver. Combiner is unnecessary because I was planning to do all in the reducer. Partitioner will distribute the mapper output based on key, reducer will create the records based on key and the create desire output for each of them. This is the following chart for this specific task:



2. Provide the full source code you used for this assignment. The source code must match the design you used for the first task.

You can use free SFTP software such as FileZilla to connect to your virtual machine and transfer the files in and out between your local machine and the virtual machine. After pulling out the source codes, you can open them on your local machine and organise them into the word document or text document for submitting to Turnitin. You don't need to report the codes in this template.

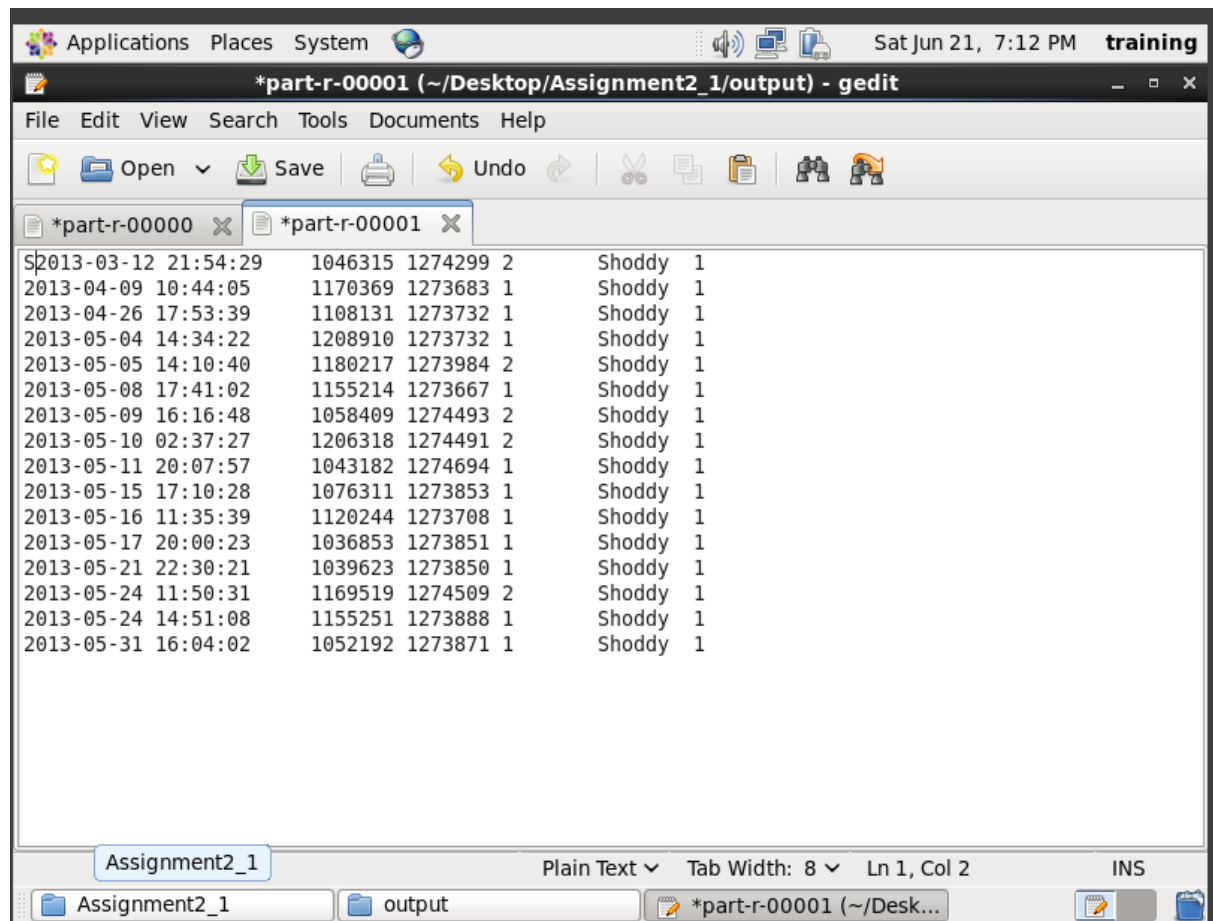
3. Show the screenshot of all locations in your result where the sentence "Item was defective" appears in the sources.



The screenshot shows a gedit text editor window titled '*part-r-00000 (~/Desktop/Assignment2_1/output) - gedit'. The window displays a CSV file with 20 rows of data. The data is as follows:

Timestamp	ID1	ID2	Count	Comment	Count
2012-11-19 23:10:40	1066812	12753686	1	Item was defective	1
2013-03-30 01:24:22	1186315	1274289	2	Item was defective	1
2013-04-08 04:03:22	1177098	1273761	1	Item was defective	1
2013-04-19 16:44:33	1131898	1273671	1	Item was defective	1
2013-04-26 15:12:08	1043182	1274055	1	Item was defective	1
2013-04-30 10:26:53	1119123	1273752	1	Item was defective	1
2013-05-01 21:54:08	1160794	1273880	1	Item was defective	1
2013-05-03 12:17:00	1173666	1273658	1	Item was defective	1
2013-05-03 14:29:51	1079543	1273968	2	Item was defective	1
2013-05-04 08:43:17	1210333	1273997	2	Item was defective	1
2013-05-08 18:23:03	1187533	1274709	2	Item was defective	1
2013-05-12 19:56:33	1136286	1273687	1	Item was defective	1
2013-05-15 14:00:33	1241716	1274359	2	Item was defective	1
2013-05-18 22:56:16	1154712	1273757	1	Item was defective	1
2013-05-20 14:44:40	1208855	1273708	1	Item was defective	1
2013-05-26 06:42:33	1043182	1273913	1	Item was defective	1
2013-05-26 15:51:43	1116304	1273875	1	Item was defective	1
2013-05-28 07:04:00	1238694	1274292	2	Item was defective	1
2013-05-31 12:18:55	1215628	1273662	1	Item was defective	1

4. Show the screenshot of all locations in your result where the comment with only a single word, "Shoddy", appears in the sources.



• Part 2

- Find a way with tools taught in this unit to check whether the language_code attribute in the source data is unique. **Note that you cannot eyeball the result. The result should be found by combining valid tools in a sequence.**

- 1.1. List the sequence of tools you used for finding the answer:

Ans: MapReduce is most suitable for this

- 1.2. Give a short explanation of which tool did what.

Ans: run mapper to read the desired column, create a conditional statement in reducer in my work I set if there is exactly 1 language code then print out result

- 1.3. List the code/command/statement and the outcome screenshot of the step in the sequence: (Note that only meaningful screenshot is required. For example, the screenshot of the MapReduce execution result is necessary, but the screenshot of the outcome from the "cd .." command is not meaningful)

[Phase 1 – compile java file to create class file]

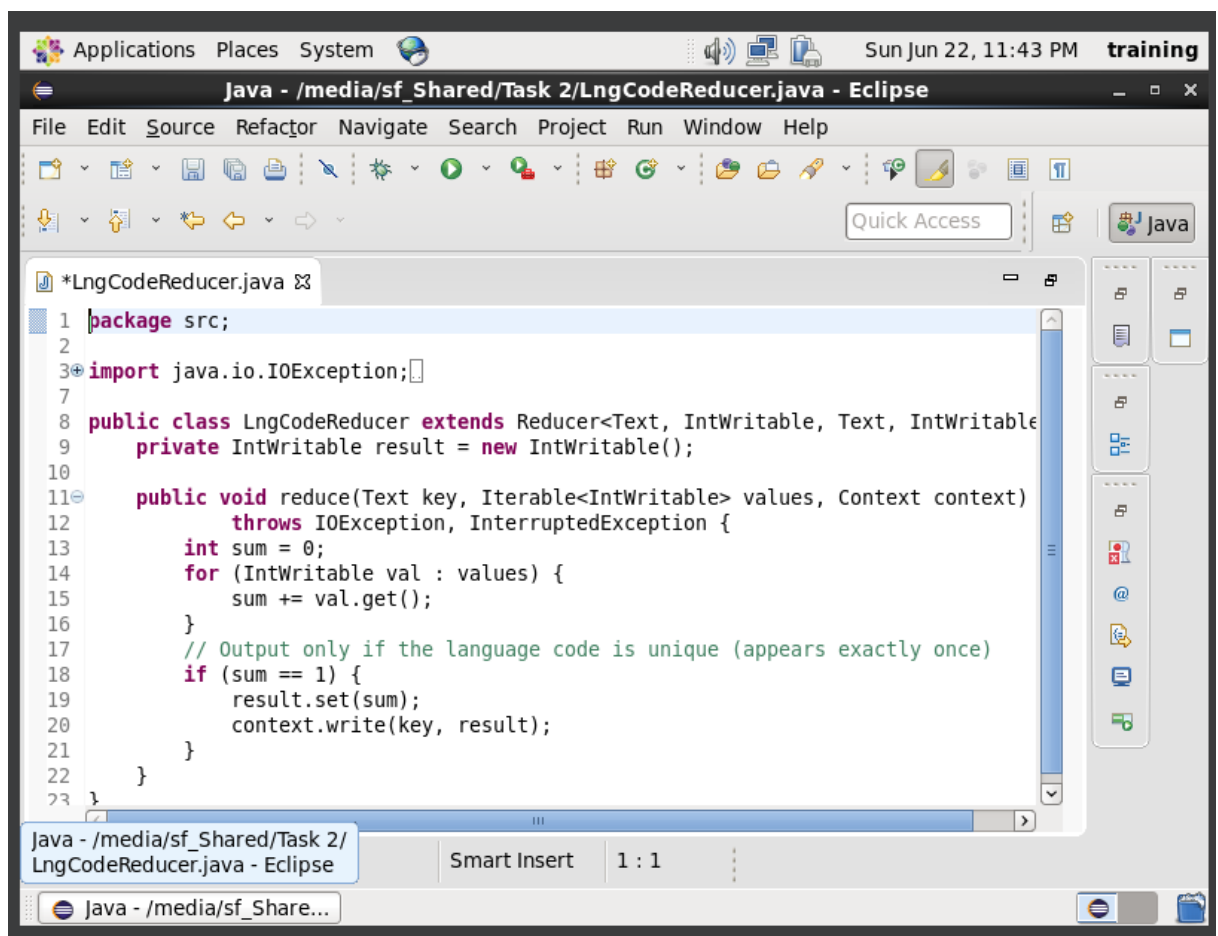
Change to the desired directory and cmd : `javac classpath `hadoop classpath``

Outcome:

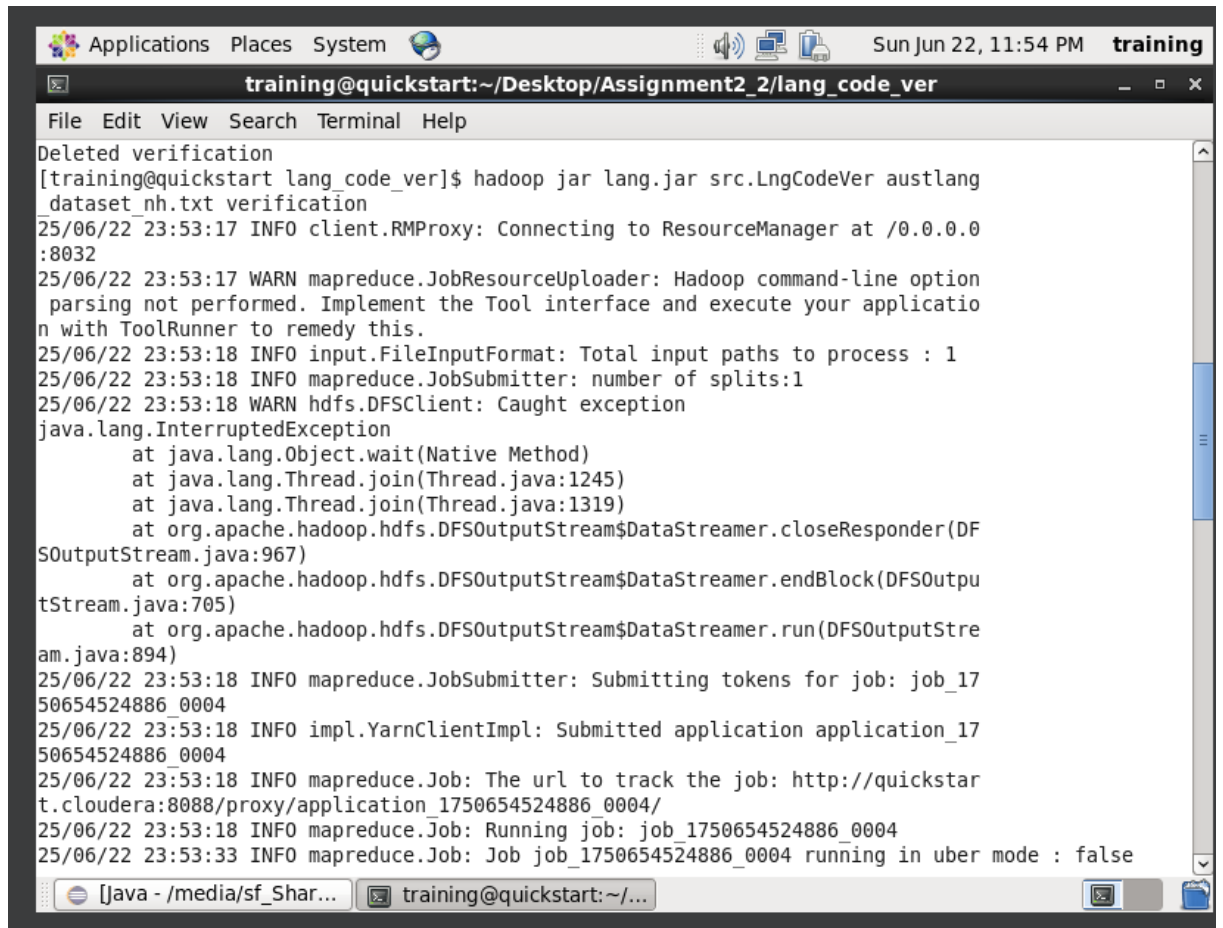
[Phase 2 – run mapreduce to see the result, if there is nothing then the lang_code attribute is not unique this is declare in the reduce file]

Command : `jar cvf lang.jar src/*.class`

Hadoop jar lang.jar src.LngCodeVer austlang_dataset_nh.txt verification



```
1 package src;
2
3 import java.io.IOException;
4
5 public class LngCodeReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
6     private IntWritable result = new IntWritable();
7
8     public void reduce(Text key, Iterable<IntWritable> values, Context context)
9         throws IOException, InterruptedException {
10         int sum = 0;
11         for (IntWritable val : values) {
12             sum += val.get();
13         }
14         // Output only if the language code is unique (appears exactly once)
15         if (sum == 1) {
16             result.set(sum);
17             context.write(key, result);
18         }
19     }
20 }
```

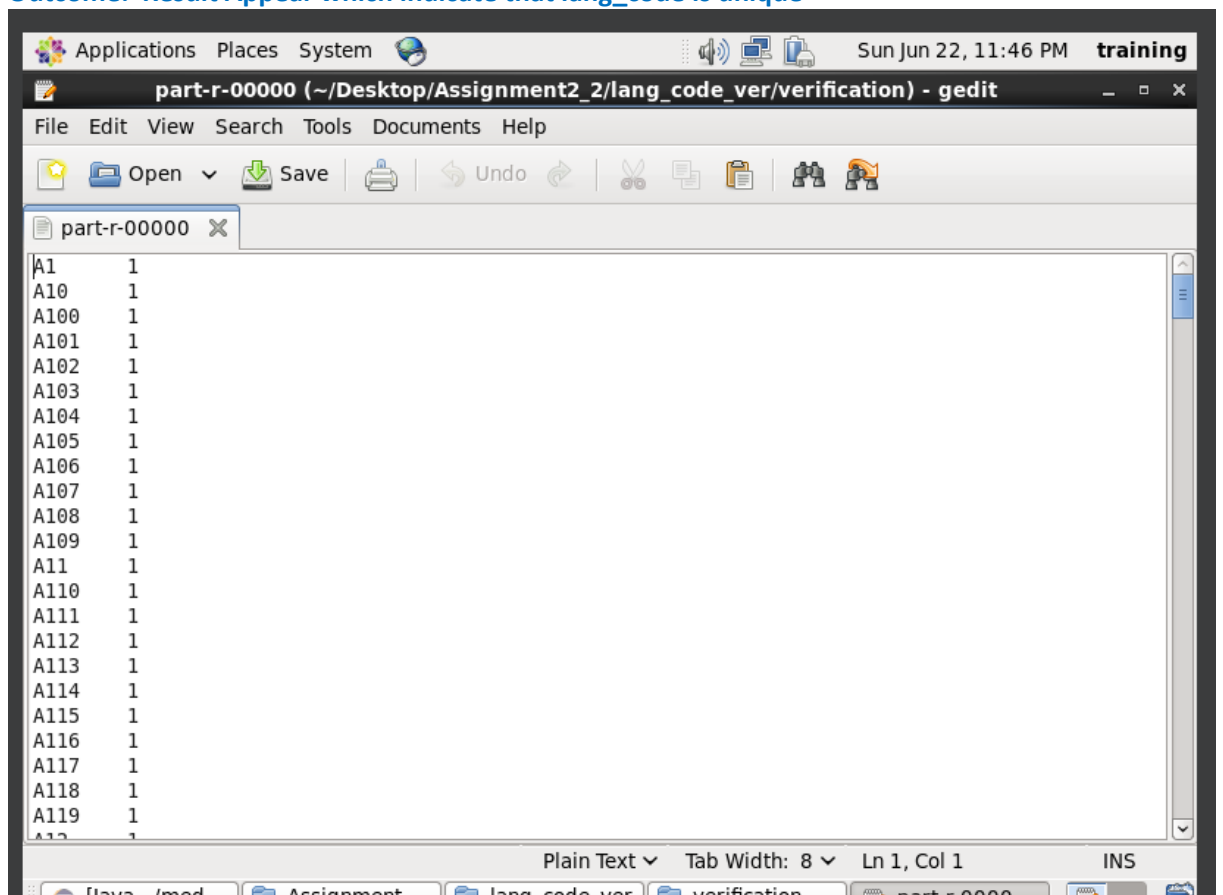


The screenshot shows a terminal window titled "training@quickstart:~/Desktop/Assignment2_2/lang_code_ver". The user has executed the command `hadoop jar lang.jar src.LngCodeVer austlang_dataset_nh.txt verification`. The output shows Hadoop logs, including a warning about command-line option parsing and a successful job submission. The job ID is `job_1750654524886_0004`.

```

Deleted verification
[training@quickstart lang_code_ver]$ hadoop jar lang.jar src.LngCodeVer austlang_dataset_nh.txt verification
25/06/22 23:53:17 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
25/06/22 23:53:17 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
25/06/22 23:53:18 INFO input.FileInputFormat: Total input paths to process : 1
25/06/22 23:53:18 INFO mapreduce.JobSubmitter: number of splits:1
25/06/22 23:53:18 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedExceptio
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1245)
    at java.lang.Thread.join(Thread.java:1319)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
25/06/22 23:53:18 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1750654524886_0004
25/06/22 23:53:18 INFO impl.YarnClientImpl: Submitted application application_1750654524886_0004
25/06/22 23:53:18 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1750654524886_0004/
25/06/22 23:53:18 INFO mapreduce.Job: Running job: job_1750654524886_0004
25/06/22 23:53:33 INFO mapreduce.Job: Job job_1750654524886_0004 running in uber mode : false
  
```

Outcome: Result Appear which indicate that lang_code is unique



The screenshot shows a gedit text editor window titled "part-r-00000 (~/Desktop/Assignment2_2/lang_code_ver/verification) - gedit". The file contains a list of unique language codes, each followed by the number 1, indicating their frequency.

```

part-r-00000
A1 1
A10 1
A100 1
A101 1
A102 1
A103 1
A104 1
A105 1
A106 1
A107 1
A108 1
A109 1
A11 1
A110 1
A111 1
A112 1
A113 1
A114 1
A115 1
A116 1
A117 1
A118 1
A119 1
A12 1
  
```


2. Find a way with tools taught in this unit to list the unique values for all entities (tables with the solid boundary) except lng_id.

2.1. What is the best tool of choice to perform this task?

Ans: MapReduce is the best tool for this task

2.2. Assume you choose MapReduce to be the tool. Should it be the Map-only design or the complete MapReduce with mapper and reducer involved in the process?

Ans: Map only is bad because we did not process the data we mapping. We need a reducer to match the data that we mapping to something meaningful like key.

2.3. Assume you choose MapReduce to be the tool. Does the count in the output matter for preparing the data for creating the entities?

Ans: No we want to count for each entity in the dataset instead of sum them up. We only count them in total if requested because it's simply not a best practice.

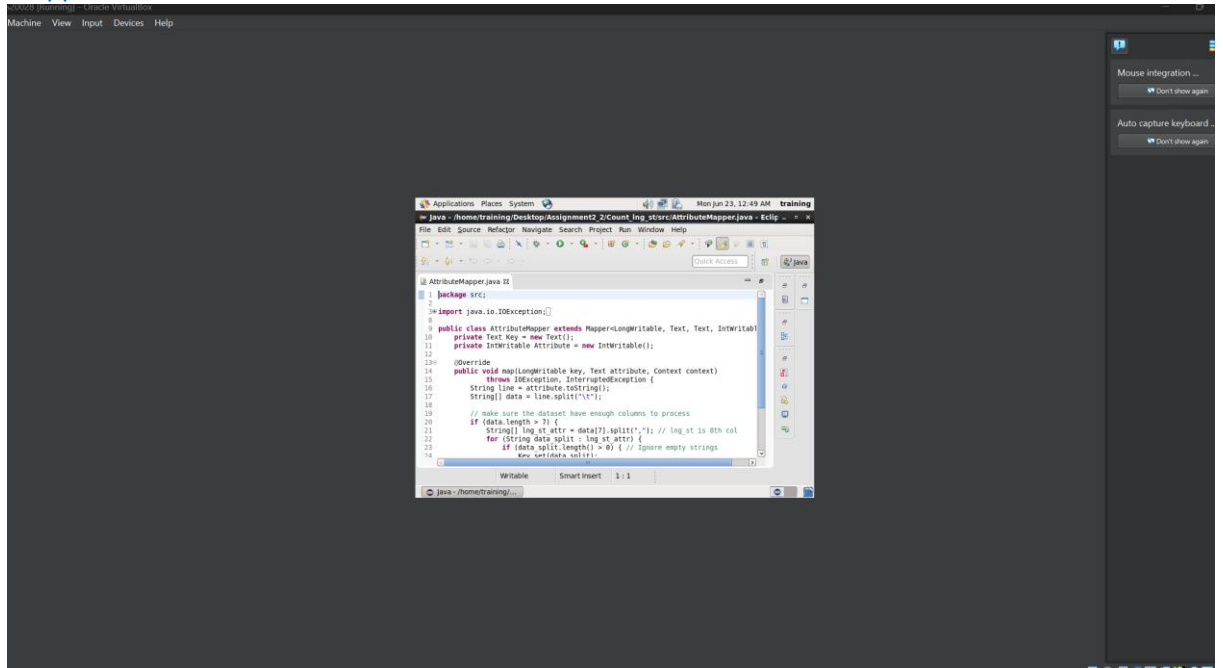
2.4. List the code/command/statement and the outcome screenshot of preparing the data for lng_st.

Ans:

Command

-the same with task 1.3 since I run MapReduce.

Mapper



```

1  package src;
2
3  import java.io.IOException;
4
5  public class AttributeMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
6      private Text key = new Text();
7      private IntWritable attribute = new IntWritable();
8
9      @Override
10     public void map(LongWritable key, Text attribute, Context context)
11         throws IOException, InterruptedException {
12         String line = attribute.toString();
13         String[] data = line.split("\t");
14
15         // make sure the dataset have enough columns to process
16         if (data.length > 7) {
17             String lng_st_attr = data[7].split("\t"); // lng_st is 8th col
18             for (String data_split : lng_st_attr) {
19                 if (data_split.length() > 0) { // ignore empty strings
20                     key.set(data_split);
21                     attribute.set(1);
22                 }
23             }
24         }
25     }
26 }
  
```

Reducer

boundary) for MySQL and Hive.

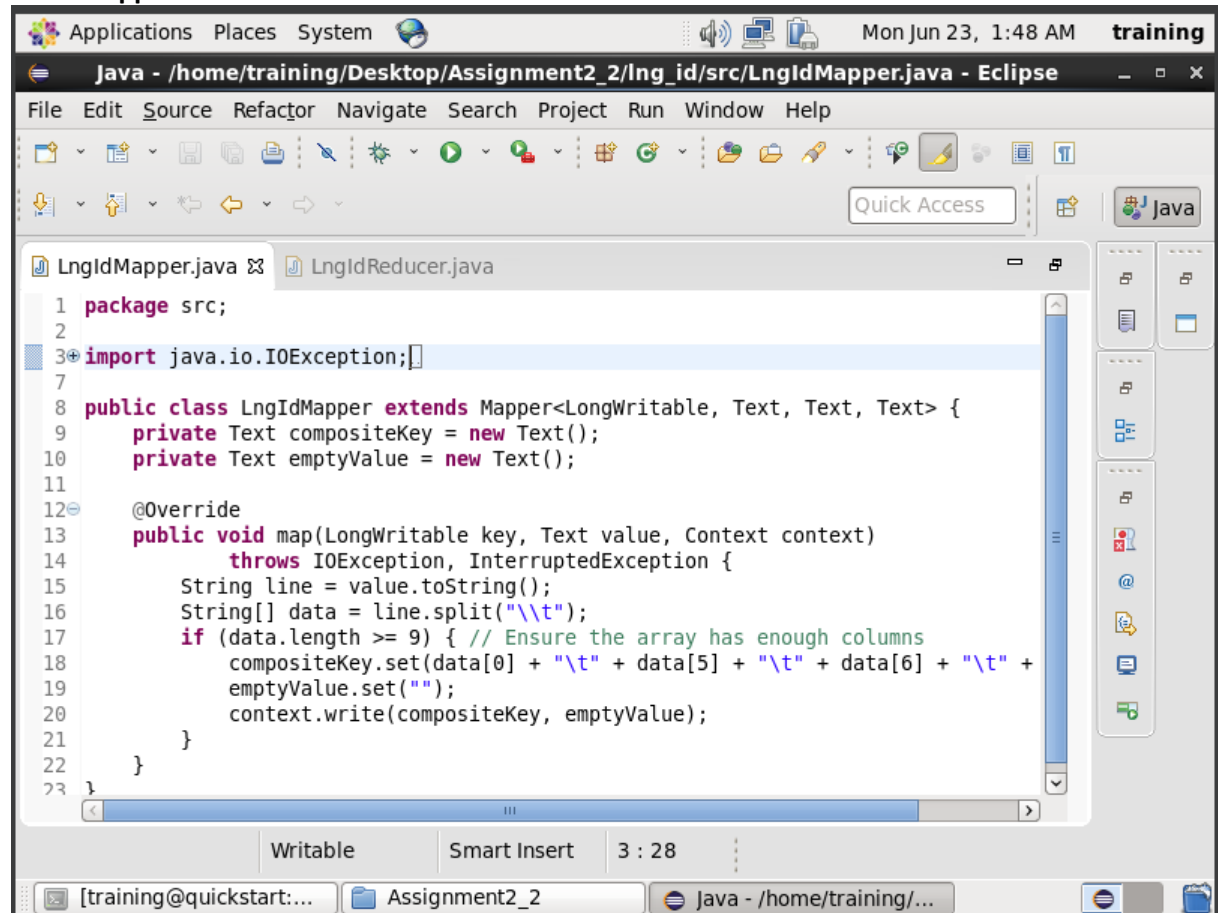
3.1. Assume your tool of choice is MapReduce for this task. Will this be a Map-only job or a complete MapReduce job for preparing the desired data?

Ans: I think it still a complete MapReduce job since I don't really expected to not using a reducer because we need to assign keys.

3.2. List the code/command/statement and the outcome screenshot of preparing the data for lng_id.

Ans: Based on the ERD provided I believe it's column 1,6,7 and 9 which is lng_code, a_lng_lat, a_lng_lng and lng_uri

Code mapper :



```

1 package src;
2
3 import java.io.IOException;
4
5
6
7
8 public class LngIdMapper extends Mapper<LongWritable, Text, Text, Text> {
9     private Text compositeKey = new Text();
10    private Text emptyValue = new Text();
11
12    @Override
13    public void map(LongWritable key, Text value, Context context)
14        throws IOException, InterruptedException {
15        String line = value.toString();
16        String[] data = line.split("\\t");
17        if (data.length >= 9) { // Ensure the array has enough columns
18            compositeKey.set(data[0] + "\\t" + data[5] + "\\t" + data[6] + "\\t" +
19                emptyValue.set("");
20            context.write(compositeKey, emptyValue);
21        }
22    }
23 }
  
```

Code Reducer:

The screenshot shows the Eclipse IDE interface. The title bar indicates the file is `Java - /home/training/Desktop/Assignment2_2/Lng_id/src/LngIdReducer.java - Eclipse`. The menu bar includes File, Edit, Source, Refactor, Navigate, Search, Project, Run, Window, and Help. The toolbar contains various icons for file operations and development. The main editor window displays the following Java code:

```

1 package src;
2
3 import java.io.IOException;
4
5
6
7
8 public class LngIdReducer extends Reducer<Text, IntWritable, Text, Text> {
9     private Text emptyValue = new Text();
10
11     @Override
12     public void reduce(Text key, Iterable<IntWritable> values, Context context)
13         throws IOException, InterruptedException {
14         emptyValue.set("");
15         context.write(key, emptyValue);
16     }
17 }

```

The bottom status bar shows the current file is `Assignment2_2` and the cursor is at line 1, column 1.

Output:

The screenshot shows a terminal window titled `part-r-00000 (~/Desktop/Assignment2_2/Lng_id/Lng_id_result) - gedit`. The terminal displays the output of a MapReduce job, showing a list of keys and values. The output is as follows:

```

A1 -32.39094519 118.7550827 https://collection.aiatsis.gov.au/austlang/language/a1
A10 https://collection.aiatsis.gov.au/austlang/language/a10
A100 https://collection.aiatsis.gov.au/austlang/language/a100
A101 -28.1355411 114.7644712 https://collection.aiatsis.gov.au/austlang/language/
a101
A102 -29.46595664 127.7662611 https://collection.aiatsis.gov.au/austlang/language/
a102
A103 -29.86593028 122.698712 https://collection.aiatsis.gov.au/austlang/language/
a103
A104 https://collection.aiatsis.gov.au/austlang/language/a104
A105 https://collection.aiatsis.gov.au/austlang/language/a105
A106 https://collection.aiatsis.gov.au/austlang/language/a106
A107 https://collection.aiatsis.gov.au/austlang/language/a107
A108 https://collection.aiatsis.gov.au/austlang/language/a108
A109 https://collection.aiatsis.gov.au/austlang/language/a109
A11 -29.231154 122.7229745 https://collection.aiatsis.gov.au/austlang/language/a11
A110 https://collection.aiatsis.gov.au/austlang/language/a110
A111 https://collection.aiatsis.gov.au/austlang/language/a111
A112 https://collection.aiatsis.gov.au/austlang/language/a112
A113 https://collection.aiatsis.gov.au/austlang/language/a113
A114 https://collection.aiatsis.gov.au/austlang/language/a114
A115 https://collection.aiatsis.gov.au/austlang/language/a115
A116 https://collection.aiatsis.gov.au/austlang/language/a116
A117 https://collection.aiatsis.gov.au/austlang/language/a117

```

The terminal window also shows the file `Assignment2_2` in the background.

4. Find a way with tools taught in this unit to prepare the data for all weak entities (tables with the dashed boundary) for MySQL and Hive.

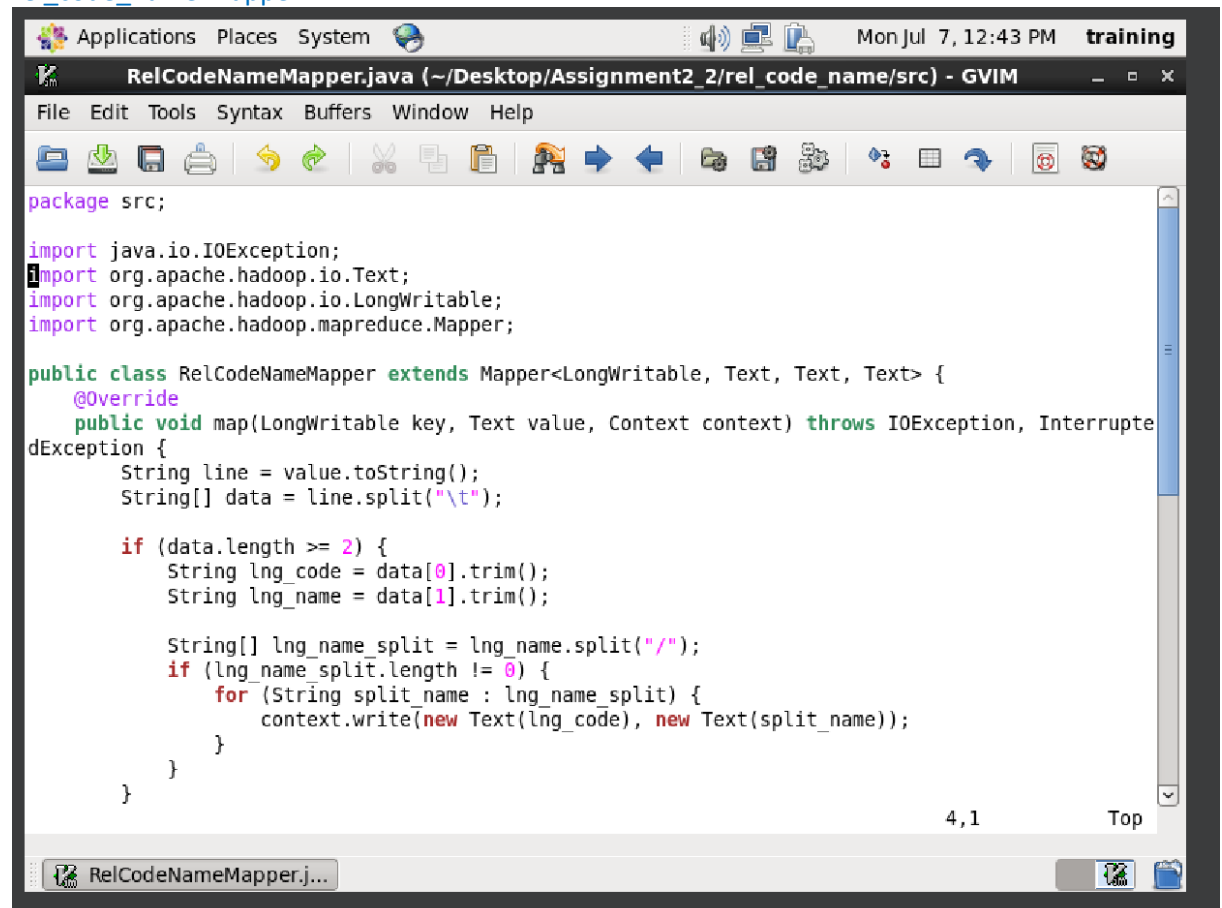
- 4.1. Which tool would be the best choice for handling this task?

Ans: MapReduce is best suit for this task since I only learn this till Week 8 (Pig is week 9 and the deadline is the end of week 8)

- 4.2. List the code/statement of preparing the data for all weak entities.

Ans:

The logic behind this is remember the column position and extract column that match the ERD provided. The reducer remain the same so I only reducer once
rel_code_name Mapper :



```
package src;

import java.io.IOException;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.mapreduce.Mapper;

public class RelCodeNameMapper extends Mapper<LongWritable, Text, Text, Text> {
    @Override
    public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
        String line = value.toString();
        String[] data = line.split("\t");

        if (data.length >= 2) {
            String lng_code = data[0].trim();
            String lng_name = data[1].trim();

            String[] lng_name_split = lng_name.split("/");
            if (lng_name_split.length != 0) {
                for (String split_name : lng_name_split) {
                    context.write(new Text(lng_code), new Text(split_name));
                }
            }
        }
    }
}
```

rel_code_name Reducer:

```

package src;

import java.io.IOException;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class RelCodeNameReducer extends Reducer<Text, Text, Text, Text> {
    @Override
    public void reduce(Text key, Iterable<Text> values, Context context) throws IOException, InterruptedException {
        for (Text value : values) {
            // Emit each value on a new line for the same key
            context.write(key, value);
        }
    }
}

```

rel_code_thp Mapper:

```

package src;

import java.io.IOException;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.mapreduce.Mapper;

public class RelCodeThpMapper extends Mapper<LongWritable, Text, Text, Text> {
    private Text keyOutput = new Text();
    private Text valueOutput = new Text();

    @Override
    public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
        String line = value.toString();
        String[] data = line.split("\t");

        if (data.length >= 2) {
            String lng_code = data[0].trim();
            String lng_thp = data[4].trim();

            String[] lng_thp_split = lng_thp.split("/");
            if (lng_thp_split.length != 0) {
                for (String split_name : lng_thp_split) {
                    context.write(new Text(lng_code), new Text(split_name));
                }
            }
        }
    }
}

```

Rel_code_synonym Mapper:

```

public void map(Object key, Text value, Context context) throws IOException, InterruptedException {

    //process input
    String line = value.toString();
    if (line.startsWith("language_code")) {
        return;
    }

    // Split line
    String[] data = line.split("\t");

    // Ensure there are at least 3 fields
    if (data.length > 2) {
        String lng_code = data[0].trim();
        String[] synonyms = data[2].split(",");

        for (String synonym : synonyms) {
            String Process_Synonym = synonym.trim();
            if (!Process_Synonym.isEmpty()) { // not null
                keyOutput.set(lng_code);
                Key_2.set(Process_Synonym);
                context.write(keyOutput, Key_2);
            }
        }
    }
}

```

rel_code_synonym 16,9 80%

Assignment2_2 rel_code_synonym src RelCodeSynonym...

rel_code_thl Mapper:

```

public class RelCodeThlMapper extends Mapper<LongWritable, Text, Text, Text> {
    private Text keyOutput = new Text();
    private Text valueOutput = new Text();

    @Override
    public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
        String line = value.toString();
        String[] data = line.split("\t");

        if (data.length >= 2) {
            String lng_code = data[0].trim();
            String lng_thl = data[3].trim();

            String[] lng_thl_split = lng_thl.split("/");
            if (lng_thl_split.length != 0) {
                for (String split_name : lng_thl_split) {
                    context.write(new Text(lng_code), new Text(split_name));
                }
            }
        }
    }
}

```

rel_code_thl 10,1 85%

Assignment2_2 rel_code_thl containing 1 item

Assignment2_2 rel_code_thl src RelCodeThlMappe...

rel_code_st Mapper:

```

private Text valueOutput = new Text();

@Override
public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
    String line = value.toString();
    String[] data = line.split("\t");

    if (data.length >= 2) {
        String lng_code = data[0]; // Primary key
        String lng_st = data[7]; // Weak entity attribute

        keyOutput.set(lng_code);
        valueOutput.set(lng_st);

        context.write(keyOutput, valueOutput);
    }
}

```

4.3. Post the first-page screenshot of the rel_code_name processed result.

Ans:

```

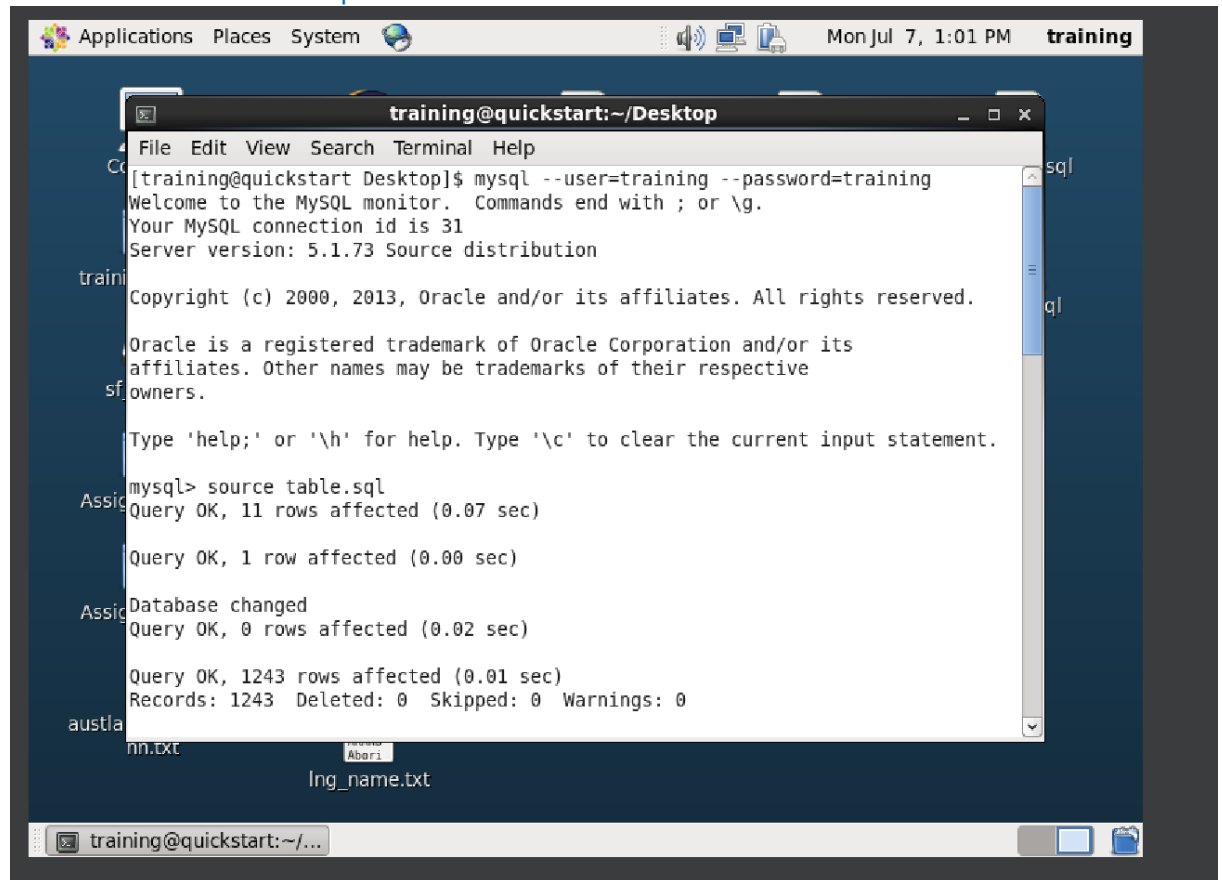
A1 Nyaki Nyaki
A1 Njaki Njaki
A10 Ngurlu
A100 Mardo
A101 Naaguja
A102 Pindiini
A103 Wangkatja
A104 Pijikala
A105 Nangu
A106 Najanaja
A107 Nawurtu
A108 Nankilakuthu
A109 Martuyitha
A11 Waljen
A110 Marlpa
A111 Windaga
A112 Ngunuru
A113 Byro Wajarri
A114 Mileura Wajarri
A115 Nharnu
A116 Nyiyaparli Padupadu
A117 Banyjima Padupadu
A118 Ngarrka
A119 Kudun

```

5. Follow the given ERDs, create tables and import data into the created tables. All tables should be put in the database entitled “indigenous”.
- 5.1. List all statements of this task for MySQL:

Ans:

Terminal command and output:

A screenshot of a Linux desktop environment. The top panel shows the application menu, places, system status, and the date/time (Mon Jul 7, 1:01 PM) and username (training). A terminal window titled 'training@quickstart: ~/Desktop' is open. It shows the execution of the 'mysql' command with user 'training' and password 'training'. The MySQL prompt 'mysql>' is shown, followed by the command 'source table.sql'. The output shows that 11 rows were affected, 1 row was affected, the database was changed, and 1243 rows were affected. The terminal also shows a file manager window with files like 'nn.txt', 'Ing_name.txt', and 'table.sql'.

```
training@quickstart:~/Desktop
File Edit View Search Terminal Help
[training@quickstart Desktop]$ mysql --user=training --password=training
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 31
Server version: 5.1.73 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

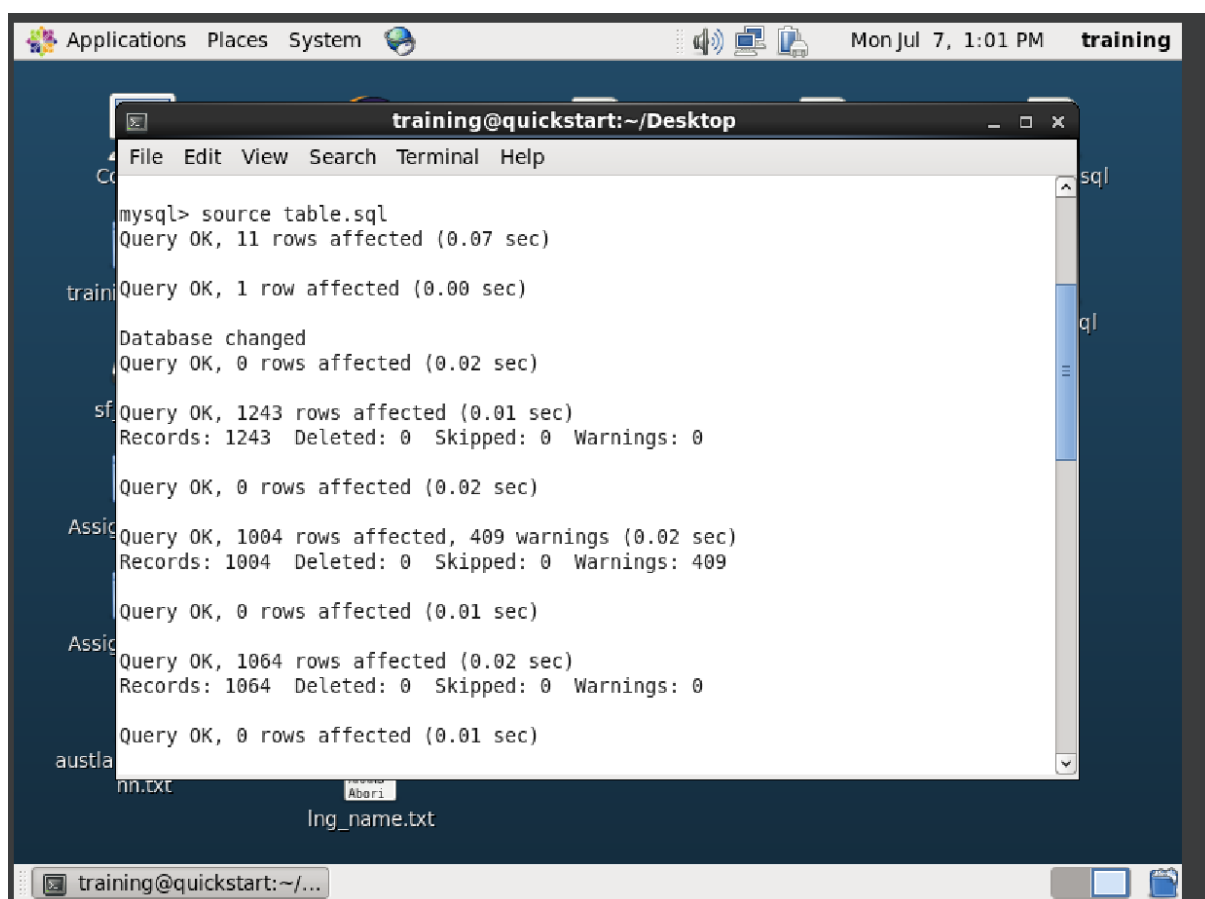
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> source table.sql
Query OK, 11 rows affected (0.07 sec)

Query OK, 1 row affected (0.00 sec)

Database changed
Query OK, 0 rows affected (0.02 sec)

Query OK, 1243 rows affected (0.01 sec)
Records: 1243  Deleted: 0  Skipped: 0  Warnings: 0
```

A screenshot of a Linux desktop environment. The top panel shows the 'Applications', 'Places', and 'System' menus, along with system status icons and the date 'Mon Jul 7, 1:01 PM'. The user is logged in as 'training'. A terminal window titled 'training@quickstart:~/Desktop' is open, displaying the following MySQL commands and output:

```
mysql> source table.sql
Query OK, 11 rows affected (0.07 sec)

Query OK, 1 row affected (0.00 sec)

Database changed
Query OK, 0 rows affected (0.02 sec)

Query OK, 1243 rows affected (0.01 sec)
Records: 1243 Deleted: 0 Skipped: 0 Warnings: 0

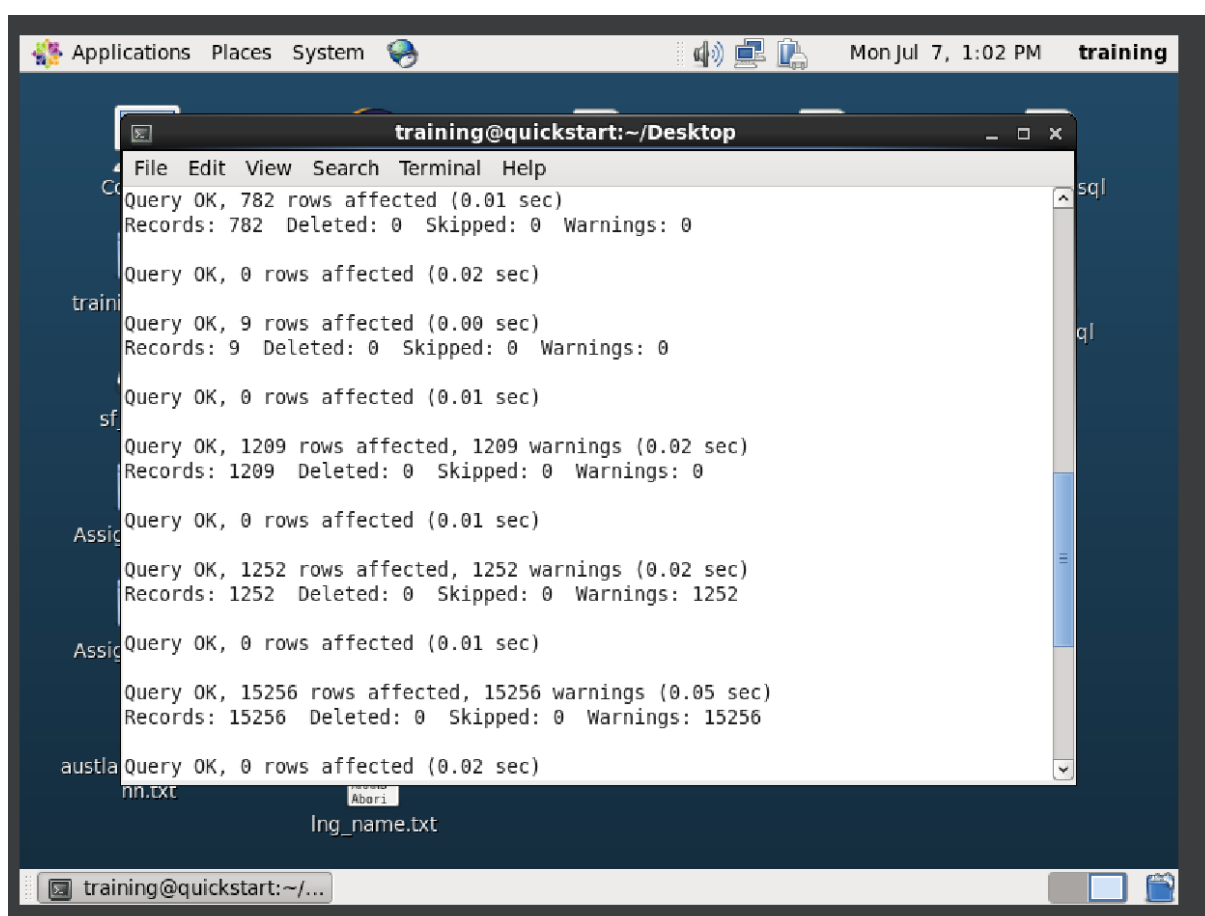
Query OK, 0 rows affected (0.02 sec)

Query OK, 1004 rows affected, 409 warnings (0.02 sec)
Records: 1004 Deleted: 0 Skipped: 0 Warnings: 409

Query OK, 0 rows affected (0.01 sec)

Query OK, 1064 rows affected (0.02 sec)
Records: 1064 Deleted: 0 Skipped: 0 Warnings: 0

Query OK, 0 rows affected (0.01 sec)
```



A screenshot of a Linux desktop environment, similar to the one above, but at 1:02 PM. The terminal window titled 'training@quickstart:~/Desktop' shows the continuation of the MySQL commands and output:

```
Query OK, 782 rows affected (0.01 sec)
Records: 782 Deleted: 0 Skipped: 0 Warnings: 0

Query OK, 0 rows affected (0.02 sec)

Query OK, 9 rows affected (0.00 sec)
Records: 9 Deleted: 0 Skipped: 0 Warnings: 0

Query OK, 0 rows affected (0.01 sec)

Query OK, 1209 rows affected, 1209 warnings (0.02 sec)
Records: 1209 Deleted: 0 Skipped: 0 Warnings: 0

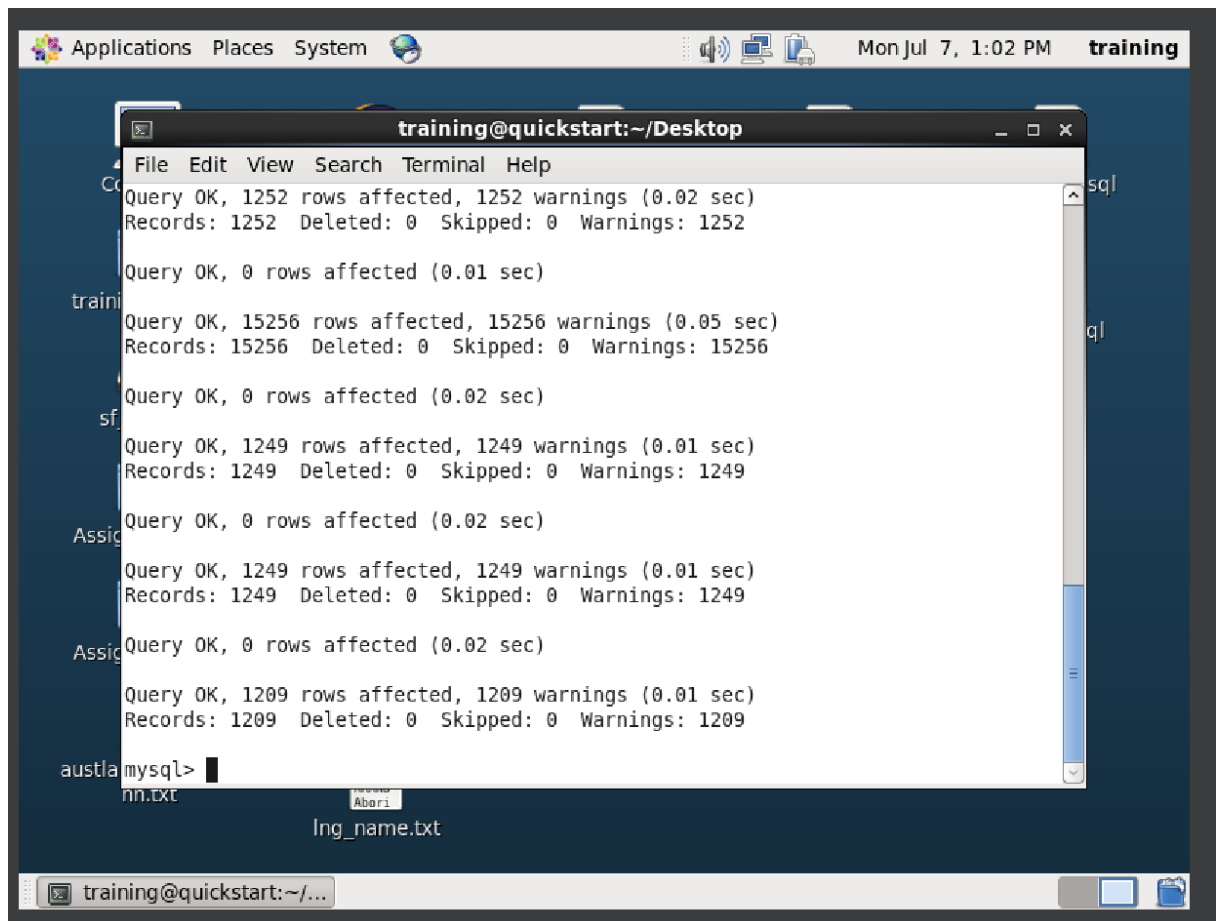
Query OK, 0 rows affected (0.01 sec)

Query OK, 1252 rows affected, 1252 warnings (0.02 sec)
Records: 1252 Deleted: 0 Skipped: 0 Warnings: 1252

Query OK, 0 rows affected (0.01 sec)

Query OK, 15256 rows affected, 15256 warnings (0.05 sec)
Records: 15256 Deleted: 0 Skipped: 0 Warnings: 15256

Query OK, 0 rows affected (0.02 sec)
```

The screenshot shows a Linux desktop environment with a terminal window titled "training@quickstart: ~/Desktop". The terminal displays the output of several MySQL queries. The queries and their results are as follows:

```
Query OK, 1252 rows affected, 1252 warnings (0.02 sec)
Records: 1252 Deleted: 0 Skipped: 0 Warnings: 1252

Query OK, 0 rows affected (0.01 sec)

Query OK, 15256 rows affected, 15256 warnings (0.05 sec)
Records: 15256 Deleted: 0 Skipped: 0 Warnings: 15256

Query OK, 0 rows affected (0.02 sec)

Query OK, 1249 rows affected, 1249 warnings (0.01 sec)
Records: 1249 Deleted: 0 Skipped: 0 Warnings: 1249

Query OK, 0 rows affected (0.02 sec)

Query OK, 1249 rows affected, 1249 warnings (0.01 sec)
Records: 1249 Deleted: 0 Skipped: 0 Warnings: 1249

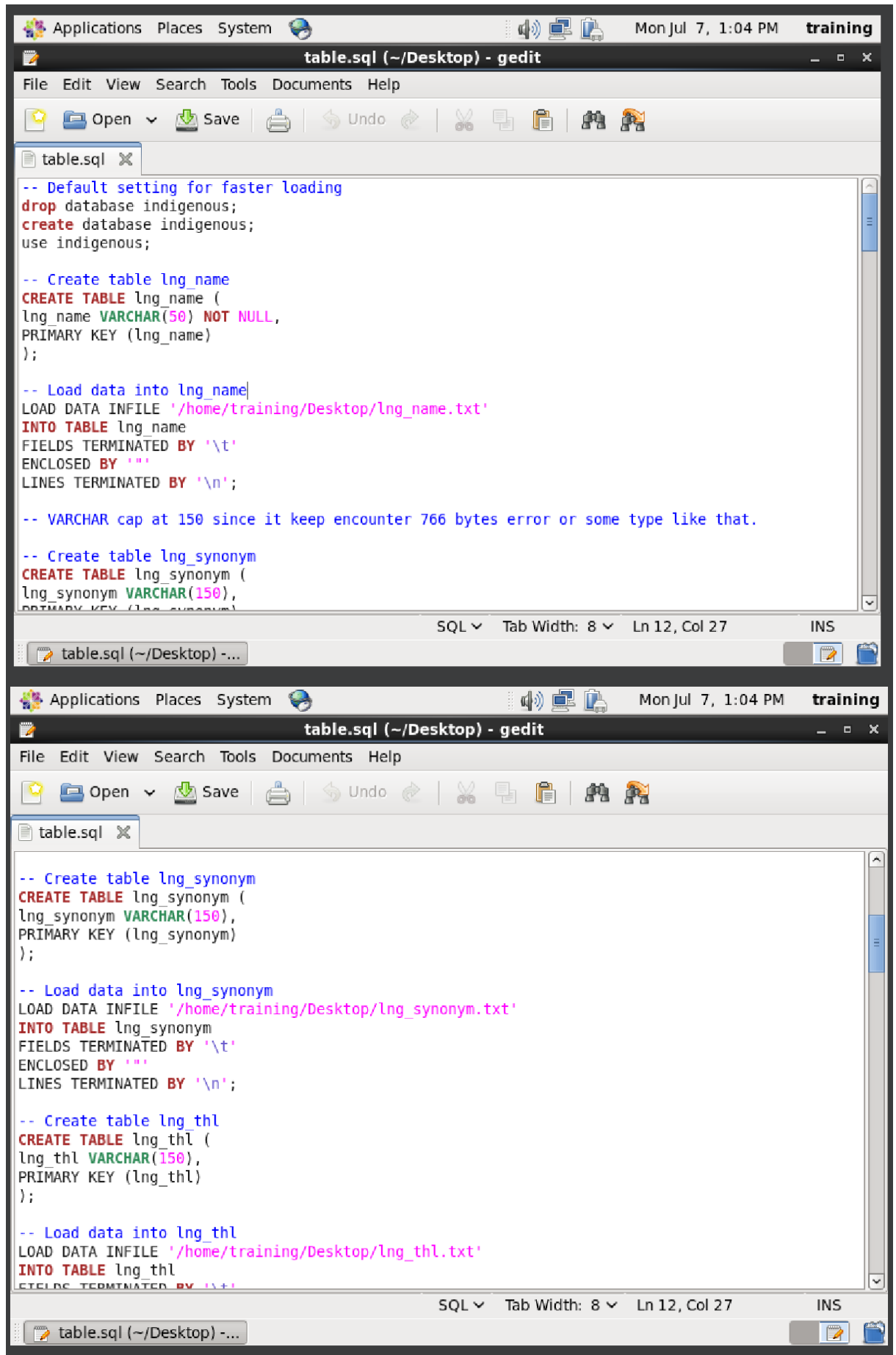
Query OK, 0 rows affected (0.02 sec)

Query OK, 1209 rows affected, 1209 warnings (0.01 sec)
Records: 1209 Deleted: 0 Skipped: 0 Warnings: 1209

mysql>
```

The terminal window also shows a file explorer in the background with files named "nn.txt", "Abori", and "Ing_name.txt". The desktop background is dark blue with a grid pattern.

Table.sql file commands:



The image displays two screenshots of a gedit editor window, showing SQL code for database setup and data loading. The window title is "table.sql (~/Desktop) - gedit". The status bar indicates "SQL", "Tab Width: 8", "Ln 12, Col 27", and "INS".

Top Screenshot:

```
-- Default setting for faster loading
drop database indigenous;
create database indigenous;
use indigenous;

-- Create table lng_name
CREATE TABLE lng_name (
  lng_name VARCHAR(50) NOT NULL,
  PRIMARY KEY (lng_name)
);

-- Load data into lng_name
LOAD DATA INFILE '/home/training/Desktop/lng_name.txt'
INTO TABLE lng_name
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

-- VARCHAR cap at 150 since it keep encounter 766 bytes error or some type like that.

-- Create table lng_synonym
CREATE TABLE lng_synonym (
  lng_synonym VARCHAR(150),
  PRIMARY KEY (lng_synonym)
);
```

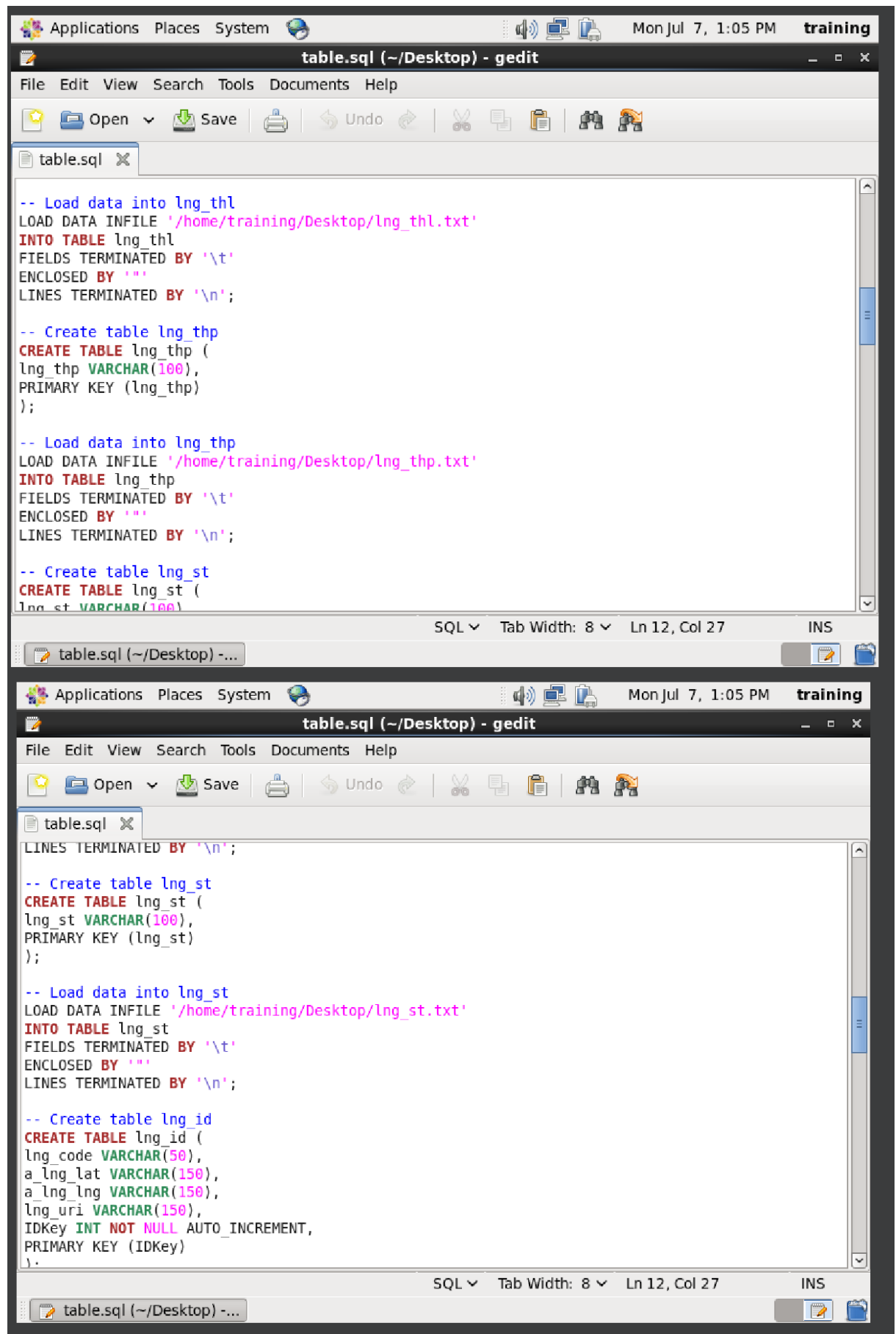
Bottom Screenshot:

```
-- Create table lng_synonym
CREATE TABLE lng_synonym (
  lng_synonym VARCHAR(150),
  PRIMARY KEY (lng_synonym)
);

-- Load data into lng_synonym
LOAD DATA INFILE '/home/training/Desktop/lng_synonym.txt'
INTO TABLE lng_synonym
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

-- Create table lng_thl
CREATE TABLE lng_thl (
  lng_thl VARCHAR(150),
  PRIMARY KEY (lng_thl)
);

-- Load data into lng_thl
LOAD DATA INFILE '/home/training/Desktop/lng_thl.txt'
INTO TABLE lng_thl
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';
```



The image shows two screenshots of a gedit editor window titled "table.sql (~/Desktop) - gedit". The window displays SQL code for creating and loading data into tables. The code is as follows:

```
-- Load data into lng_thl
LOAD DATA INFILE '/home/training/Desktop/lng_thl.txt'
INTO TABLE lng_thl
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

-- Create table lng_thp
CREATE TABLE lng_thp (
lng_thp VARCHAR(100),
PRIMARY KEY (lng_thp)
);

-- Load data into lng_thp
LOAD DATA INFILE '/home/training/Desktop/lng_thp.txt'
INTO TABLE lng_thp
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

-- Create table lng_st
CREATE TABLE lng_st (
lng_st VARCHAR(100)
);
```

The second screenshot shows the continuation of the SQL code:

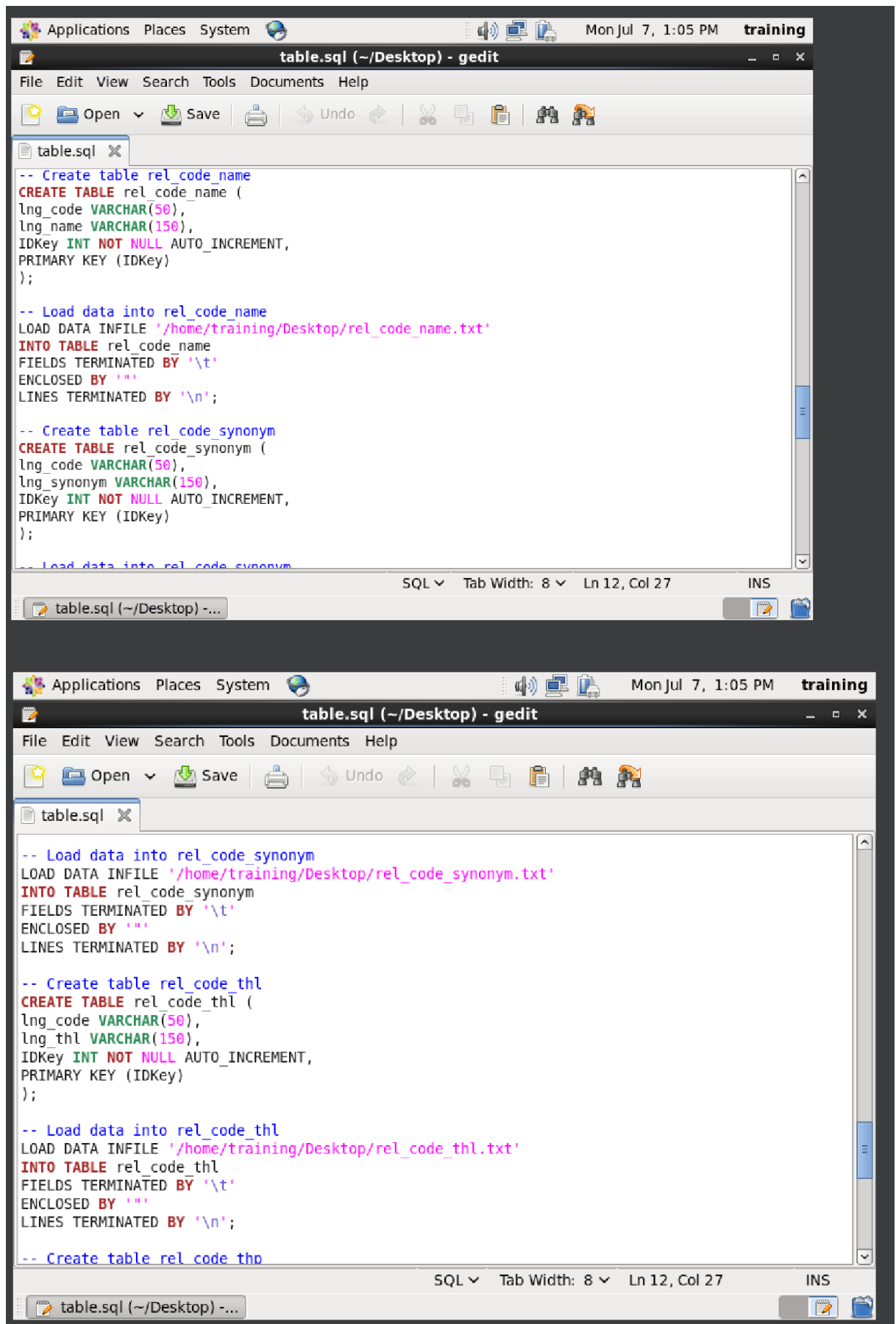
```
LINES TERMINATED BY '\n';

-- Create table lng_st
CREATE TABLE lng_st (
lng_st VARCHAR(100),
PRIMARY KEY (lng_st)
);

-- Load data into lng_st
LOAD DATA INFILE '/home/training/Desktop/lng_st.txt'
INTO TABLE lng_st
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

-- Create table lng_id
CREATE TABLE lng_id (
lng_code VARCHAR(50),
a_lng_lat VARCHAR(150),
a_lng_lng VARCHAR(150),
lng_uri VARCHAR(150),
IDKey INT NOT NULL AUTO_INCREMENT,
PRIMARY KEY (IDKey)
);
```

The status bar at the bottom of the editor shows "SQL", "Tab Width: 8", "Ln 12, Col 27", and "INS".



The image displays two screenshots of a gedit editor window titled "table.sql (~/Desktop) - gedit". The window shows SQL code for creating and loading data into tables. The top screenshot shows the first two tables, and the bottom screenshot shows the next two tables.

```
-- Create table rel_code_name
CREATE TABLE rel_code_name (
  lng_code VARCHAR(50),
  lng_name VARCHAR(150),
  IDKey INT NOT NULL AUTO_INCREMENT,
  PRIMARY KEY (IDKey)
);

-- Load data into rel_code_name
LOAD DATA INFILE '/home/training/Desktop/rel_code_name.txt'
INTO TABLE rel_code_name
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

-- Create table rel_code_synonym
CREATE TABLE rel_code_synonym (
  lng_code VARCHAR(50),
  lng_synonym VARCHAR(150),
  IDKey INT NOT NULL AUTO_INCREMENT,
  PRIMARY KEY (IDKey)
);

-- Load data into rel_code_synonym
```

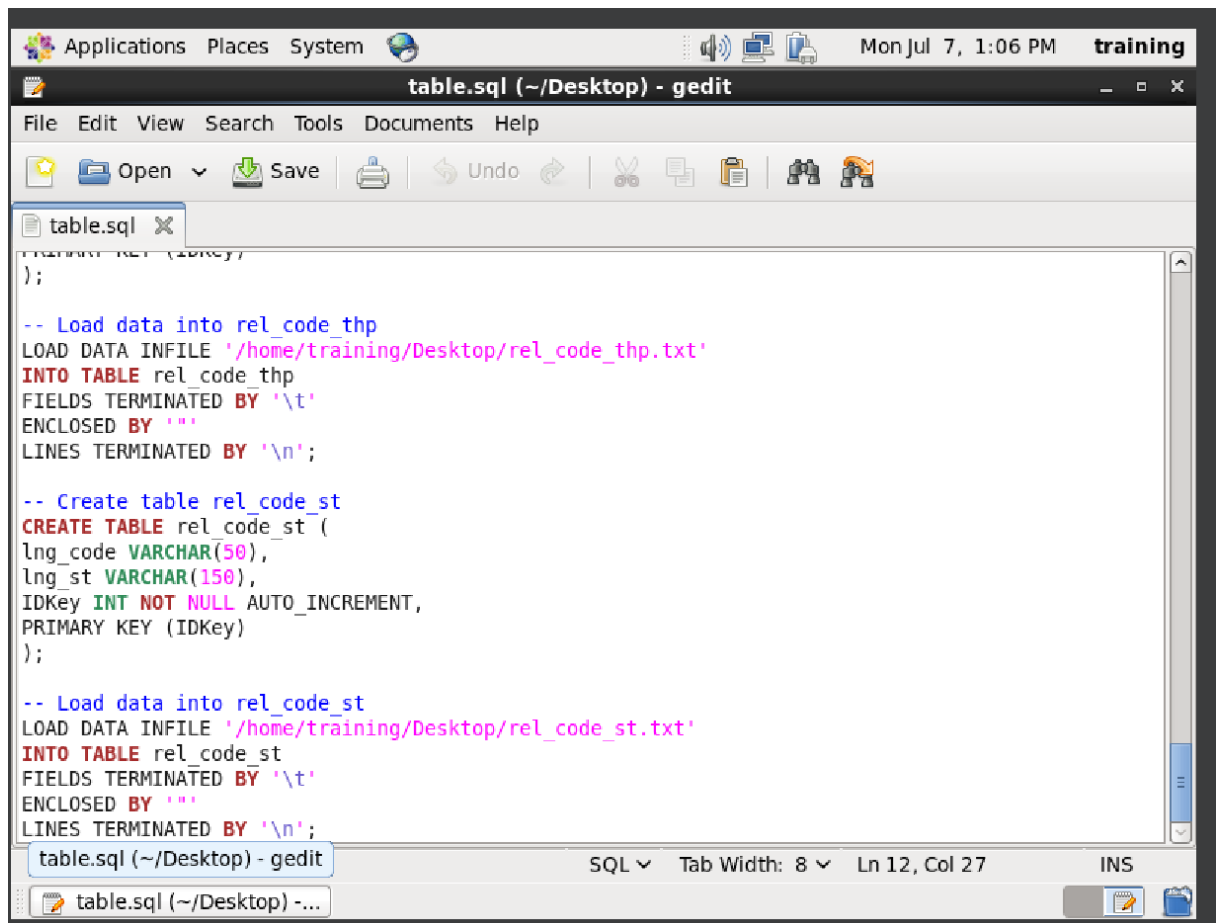
The bottom screenshot shows the next two tables:

```
-- Load data into rel_code_synonym
LOAD DATA INFILE '/home/training/Desktop/rel_code_synonym.txt'
INTO TABLE rel_code_synonym
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

-- Create table rel_code_thl
CREATE TABLE rel_code_thl (
  lng_code VARCHAR(50),
  lng_thl VARCHAR(150),
  IDKey INT NOT NULL AUTO_INCREMENT,
  PRIMARY KEY (IDKey)
);

-- Load data into rel_code_thl
LOAD DATA INFILE '/home/training/Desktop/rel_code_thl.txt'
INTO TABLE rel_code_thl
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

-- Create table rel_code_thp
```



```
table.sql (~/Desktop) - gedit
File Edit View Search Tools Documents Help
Open Save Undo
table.sql
-- Load data into rel_code_thp
LOAD DATA INFILE '/home/training/Desktop/rel_code_thp.txt'
INTO TABLE rel_code_thp
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

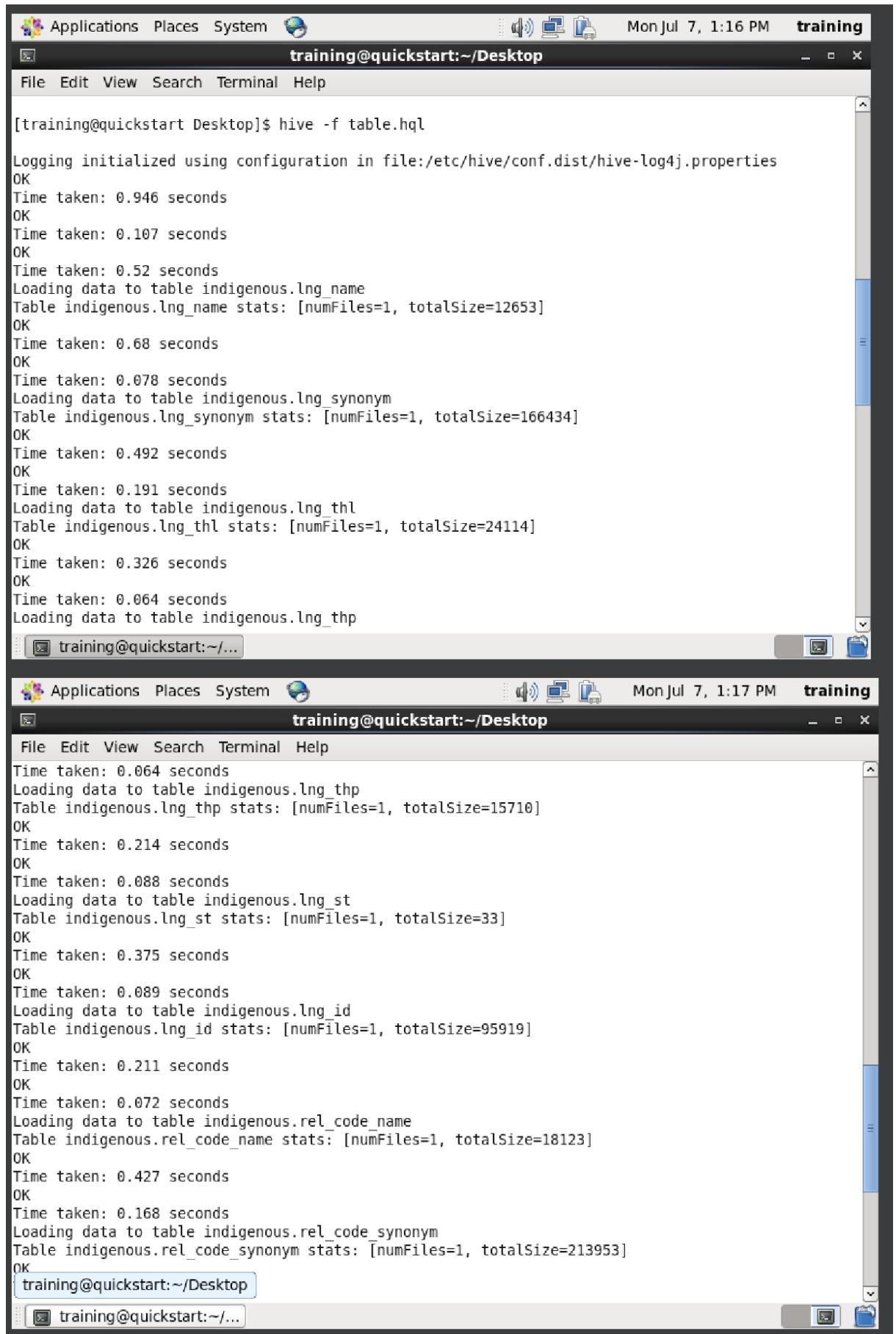
-- Create table rel_code_st
CREATE TABLE rel_code_st (
  lng_code VARCHAR(50),
  lng_st VARCHAR(150),
  IDKey INT NOT NULL AUTO_INCREMENT,
  PRIMARY KEY (IDKey)
);

-- Load data into rel_code_st
LOAD DATA INFILE '/home/training/Desktop/rel_code_st.txt'
INTO TABLE rel_code_st
FIELDS TERMINATED BY '\t'
ENCLOSED BY '"'
LINES TERMINATED BY '\n';
table.sql (~/Desktop) - gedit
SQL Tab Width: 8 Ln 12, Col 27 INS
table.sql (~/Desktop) - ...
```

5.2. List all statements of this task for Hive:

Ans:

Hive command and output:

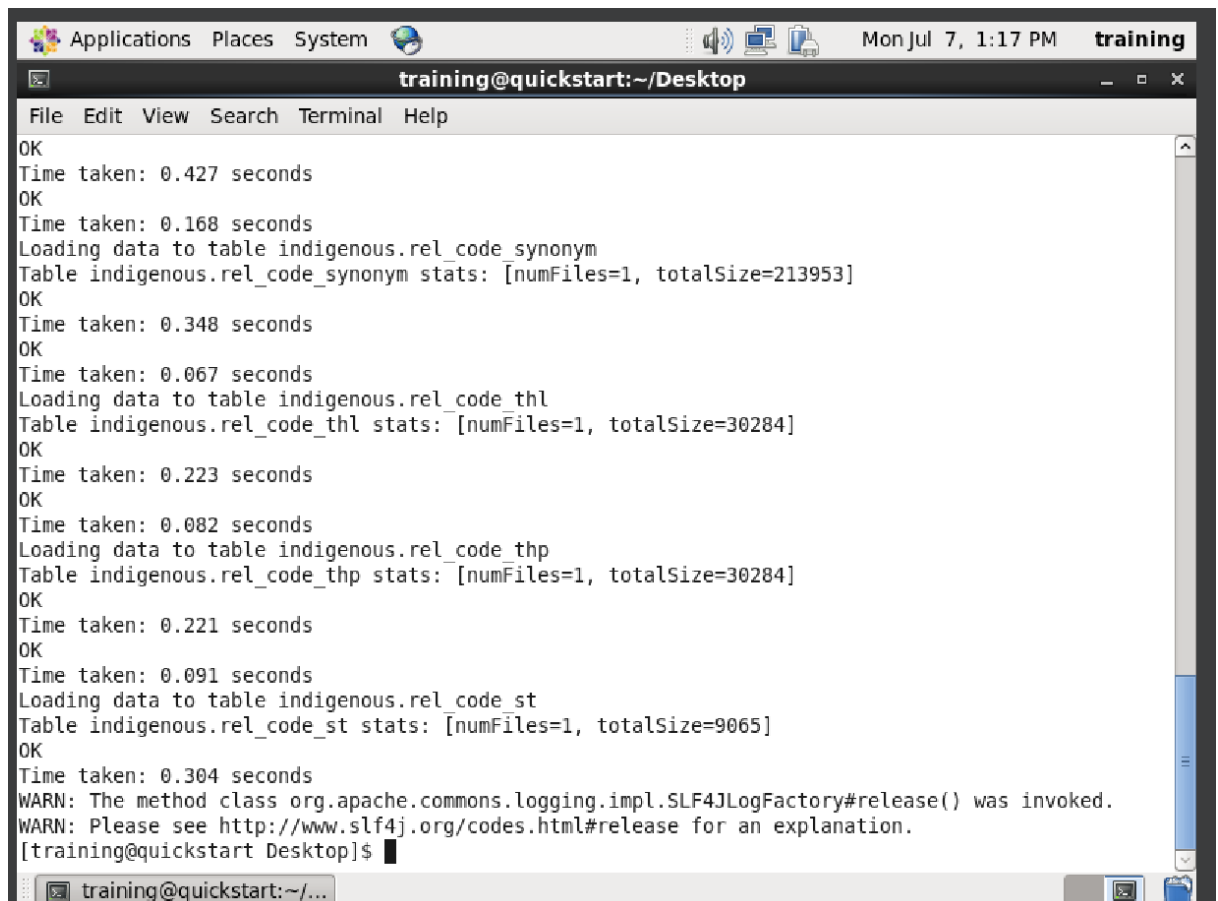


The image displays two screenshots of a Linux terminal window. The top screenshot shows the execution of the command `hive -f table.hql` and the subsequent loading of data into three tables: `indigenous.lng_name`, `indigenous.lng_synonym`, and `indigenous.lng_thl`. The bottom screenshot continues the process, showing the loading of data into `indigenous.lng_thp`, `indigenous.lng_st`, `indigenous.lng_id`, `indigenous.rel_code_name`, and `indigenous.rel_code_synonym`. Each table load operation reports the number of files and the total size of the data.

```
[training@quickstart Desktop]$ hive -f table.hql

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.properties
OK
Time taken: 0.946 seconds
OK
Time taken: 0.107 seconds
OK
Time taken: 0.52 seconds
Loading data to table indigenous.lng_name
Table indigenous.lng_name stats: [numFiles=1, totalSize=12653]
OK
Time taken: 0.68 seconds
OK
Time taken: 0.078 seconds
Loading data to table indigenous.lng_synonym
Table indigenous.lng_synonym stats: [numFiles=1, totalSize=166434]
OK
Time taken: 0.492 seconds
OK
Time taken: 0.191 seconds
Loading data to table indigenous.lng_thl
Table indigenous.lng_thl stats: [numFiles=1, totalSize=24114]
OK
Time taken: 0.326 seconds
OK
Time taken: 0.064 seconds
Loading data to table indigenous.lng_thp
training@quickstart:~/...

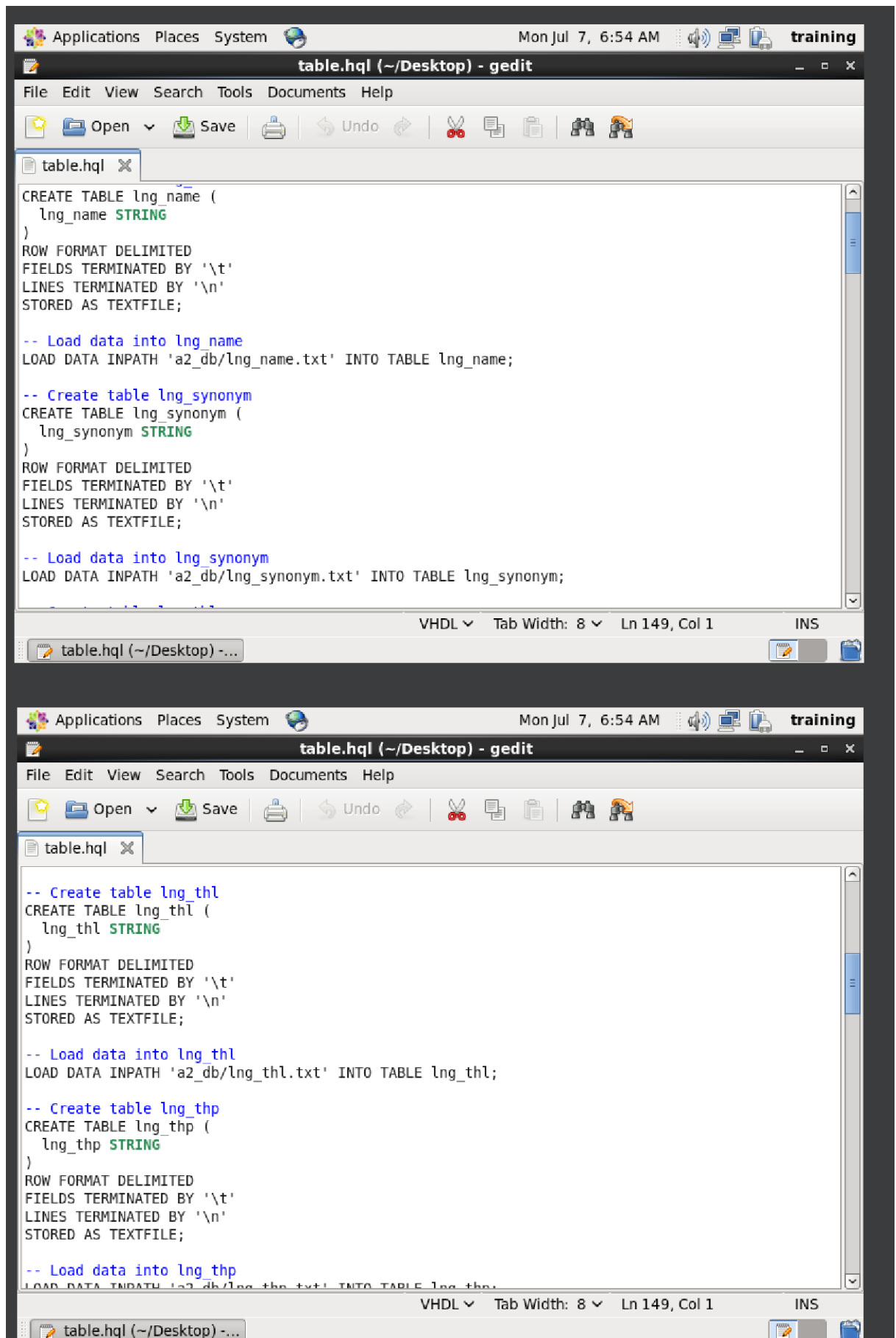
Time taken: 0.064 seconds
Loading data to table indigenous.lng_thp
Table indigenous.lng_thp stats: [numFiles=1, totalSize=15710]
OK
Time taken: 0.214 seconds
OK
Time taken: 0.088 seconds
Loading data to table indigenous.lng_st
Table indigenous.lng_st stats: [numFiles=1, totalSize=33]
OK
Time taken: 0.375 seconds
OK
Time taken: 0.089 seconds
Loading data to table indigenous.lng_id
Table indigenous.lng_id stats: [numFiles=1, totalSize=95919]
OK
Time taken: 0.211 seconds
OK
Time taken: 0.072 seconds
Loading data to table indigenous.rel_code_name
Table indigenous.rel_code_name stats: [numFiles=1, totalSize=18123]
OK
Time taken: 0.427 seconds
OK
Time taken: 0.168 seconds
Loading data to table indigenous.rel_code_synonym
Table indigenous.rel_code_synonym stats: [numFiles=1, totalSize=213953]
OK
training@quickstart:~/Desktop
training@quickstart:~/...
```



The screenshot shows a terminal window titled "training@quickstart:~/Desktop". The window contains the following text:

```
File Edit View Search Terminal Help
OK
Time taken: 0.427 seconds
OK
Time taken: 0.168 seconds
Loading data to table indigenous.rel_code_synonym
Table indigenous.rel_code_synonym stats: [numFiles=1, totalSize=213953]
OK
Time taken: 0.348 seconds
OK
Time taken: 0.067 seconds
Loading data to table indigenous.rel_code_thl
Table indigenous.rel_code_thl stats: [numFiles=1, totalSize=30284]
OK
Time taken: 0.223 seconds
OK
Time taken: 0.082 seconds
Loading data to table indigenous.rel_code_thp
Table indigenous.rel_code_thp stats: [numFiles=1, totalSize=30284]
OK
Time taken: 0.221 seconds
OK
Time taken: 0.091 seconds
Loading data to table indigenous.rel_code_st
Table indigenous.rel_code_st stats: [numFiles=1, totalSize=9065]
OK
Time taken: 0.304 seconds
WARN: The method class org.apache.commons.logging.impl.SLF4JLogFactory#release() was invoked.
WARN: Please see http://www.slf4j.org/codes.html#release for an explanation.
[training@quickstart Desktop]$
```

table.hql commands :



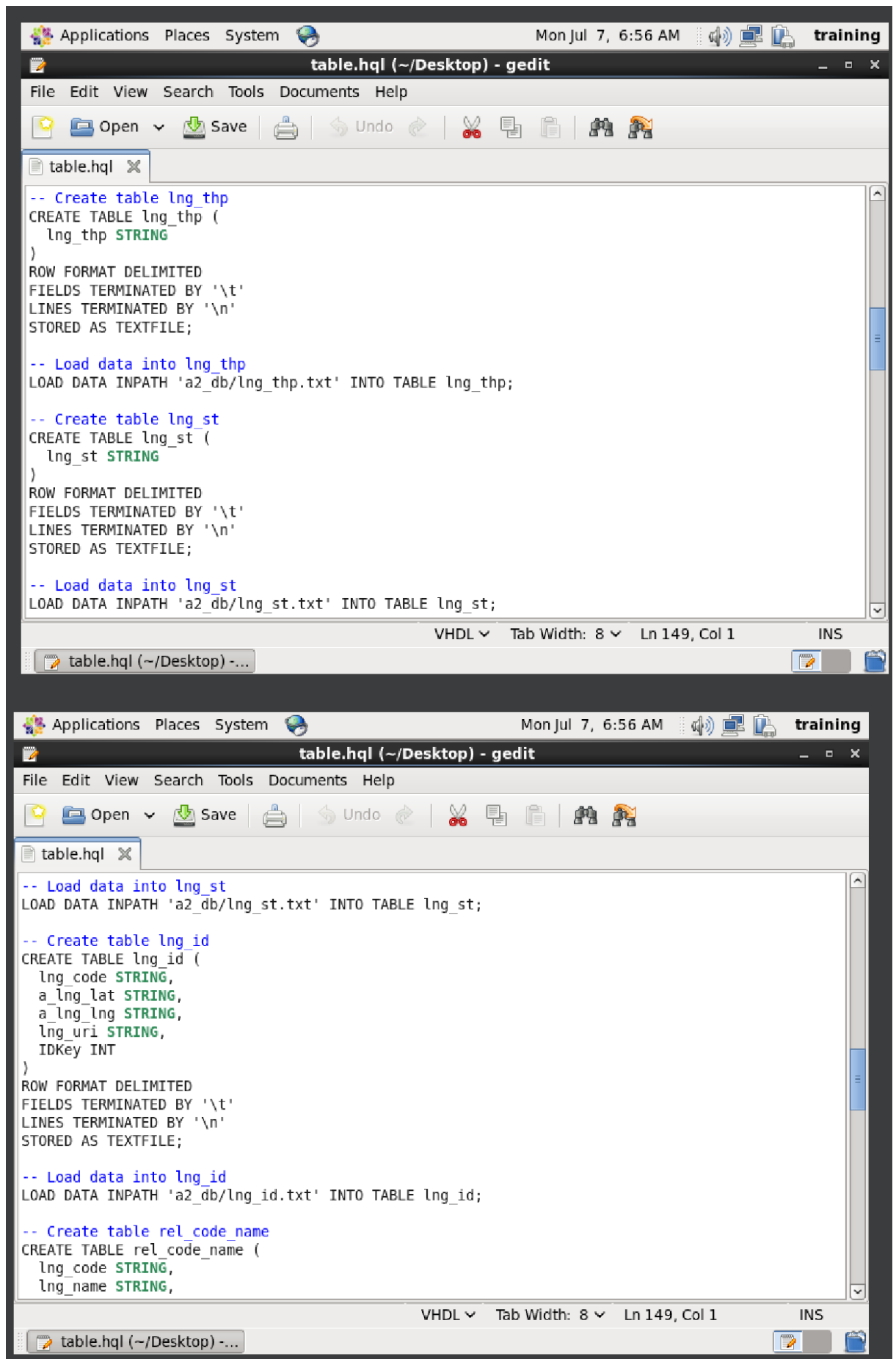
The image displays two screenshots of a gedit editor window titled "table.hql (~/Desktop) - gedit". The window shows HQL code for creating and loading data into tables. The top screenshot shows the first two tables, and the bottom screenshot shows the next two tables.

Top Screenshot:

```
CREATE TABLE lng_name (  
  lng_name STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY '\t'  
LINES TERMINATED BY '\n'  
STORED AS TEXTFILE;  
  
-- Load data into lng_name  
LOAD DATA INPATH 'a2_db/lng_name.txt' INTO TABLE lng_name;  
  
-- Create table lng_synonym  
CREATE TABLE lng_synonym (  
  lng_synonym STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY '\t'  
LINES TERMINATED BY '\n'  
STORED AS TEXTFILE;  
  
-- Load data into lng_synonym  
LOAD DATA INPATH 'a2_db/lng_synonym.txt' INTO TABLE lng_synonym;
```

Bottom Screenshot:

```
-- Create table lng_thl  
CREATE TABLE lng_thl (  
  lng_thl STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY '\t'  
LINES TERMINATED BY '\n'  
STORED AS TEXTFILE;  
  
-- Load data into lng_thl  
LOAD DATA INPATH 'a2_db/lng_thl.txt' INTO TABLE lng_thl;  
  
-- Create table lng_thp  
CREATE TABLE lng_thp (  
  lng_thp STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY '\t'  
LINES TERMINATED BY '\n'  
STORED AS TEXTFILE;  
  
-- Load data into lng_thp  
LOAD DATA INPATH 'a2_db/lng_thp.txt' INTO TABLE lng_thp;
```

The image displays two screenshots of a Gedit editor window, showing HQL code for creating and loading data into tables. The window title is "table.hql (~/Desktop) - gedit". The status bar at the bottom indicates "VHDL", "Tab Width: 8", "Ln 149, Col 1", and "INS".

Top Screenshot:

```
-- Create table lng_thp
CREATE TABLE lng_thp (
  lng_thp STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t'
LINES TERMINATED BY '\n'
STORED AS TEXTFILE;

-- Load data into lng_thp
LOAD DATA INPATH 'a2_db/lng_thp.txt' INTO TABLE lng_thp;

-- Create table lng_st
CREATE TABLE lng_st (
  lng_st STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t'
LINES TERMINATED BY '\n'
STORED AS TEXTFILE;

-- Load data into lng_st
LOAD DATA INPATH 'a2_db/lng_st.txt' INTO TABLE lng_st;
```

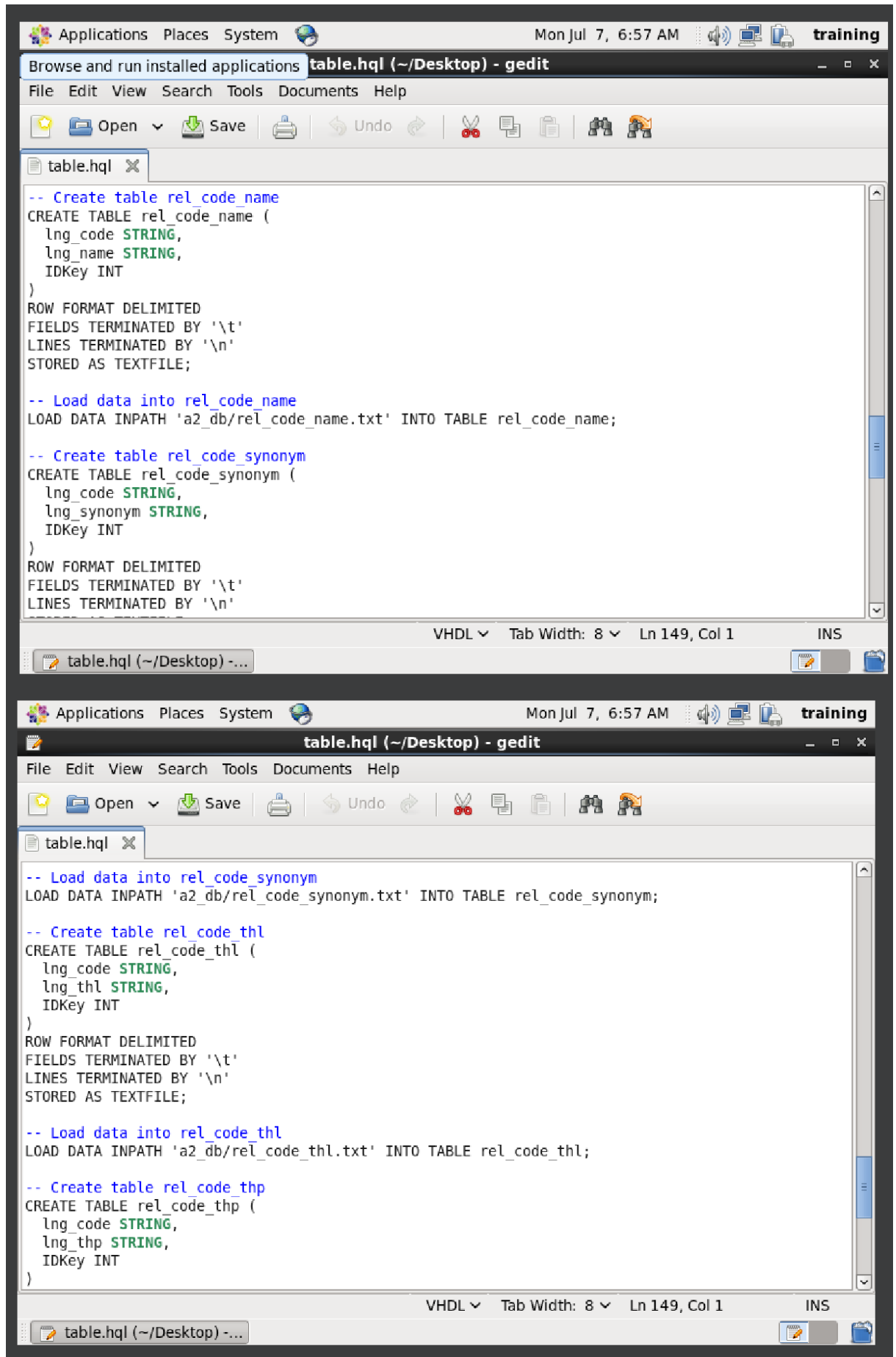
Bottom Screenshot:

```
-- Load data into lng_st
LOAD DATA INPATH 'a2_db/lng_st.txt' INTO TABLE lng_st;

-- Create table lng_id
CREATE TABLE lng_id (
  lng_code STRING,
  a_lng_lat STRING,
  a_lng_lng STRING,
  lng_uri STRING,
  IDKey INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t'
LINES TERMINATED BY '\n'
STORED AS TEXTFILE;

-- Load data into lng_id
LOAD DATA INPATH 'a2_db/lng_id.txt' INTO TABLE lng_id;

-- Create table rel_code_name
CREATE TABLE rel_code_name (
  lng_code STRING,
  lng_name STRING,
```



The image displays two screenshots of a gedit editor window, showing SQL code for creating and loading data into tables. The window title is "table.hql (~/Desktop) - gedit". The status bar indicates "VHDL", "Tab Width: 8", "Ln 149, Col 1", and "INS".

Top Screenshot:

```
-- Create table rel_code_name
CREATE TABLE rel_code_name (
  lng_code STRING,
  lng_name STRING,
  IDKey INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t'
LINES TERMINATED BY '\n'
STORED AS TEXTFILE;

-- Load data into rel_code_name
LOAD DATA INPATH 'a2_db/rel_code_name.txt' INTO TABLE rel_code_name;

-- Create table rel_code_synonym
CREATE TABLE rel_code_synonym (
  lng_code STRING,
  lng_synonym STRING,
  IDKey INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t'
LINES TERMINATED BY '\n'
```

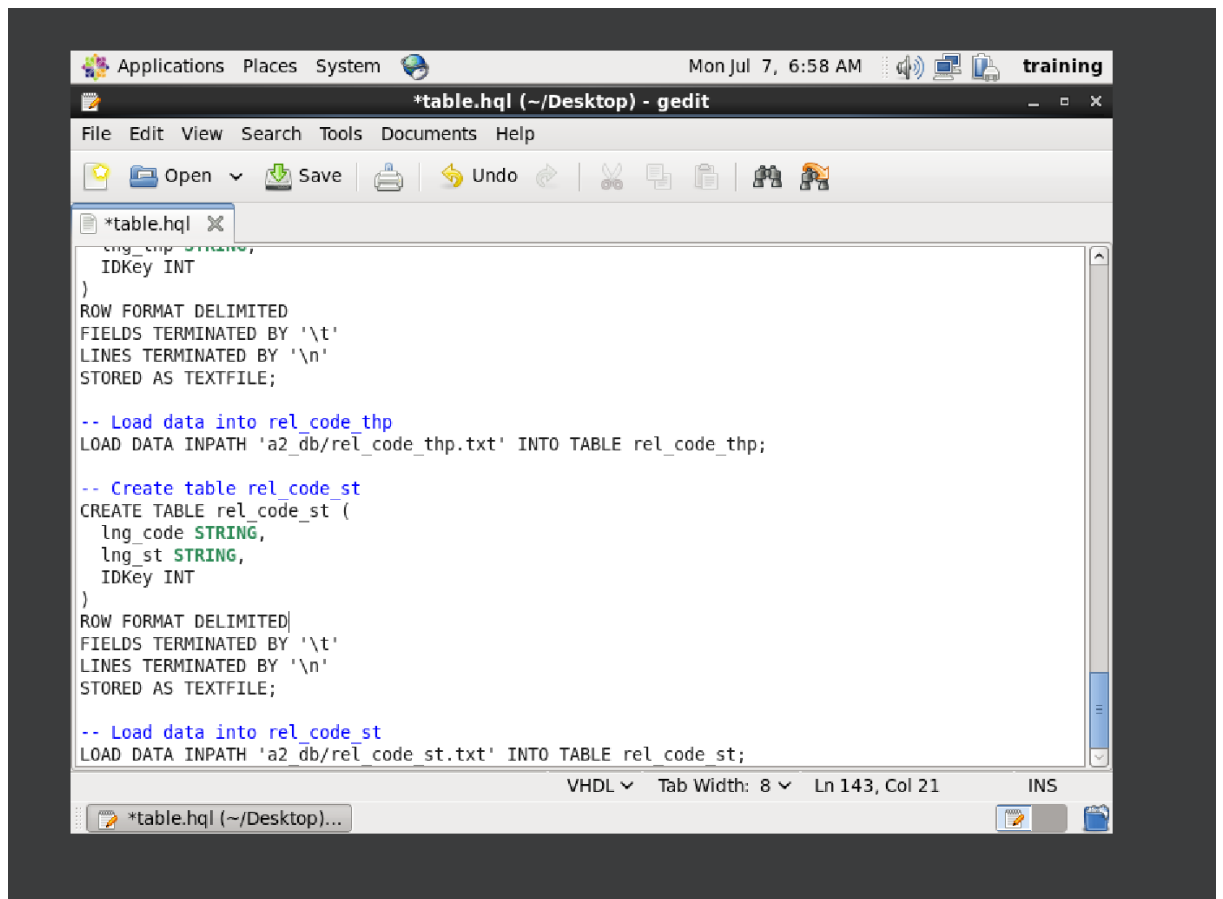
Bottom Screenshot:

```
-- Load data into rel_code_synonym
LOAD DATA INPATH 'a2_db/rel_code_synonym.txt' INTO TABLE rel_code_synonym;

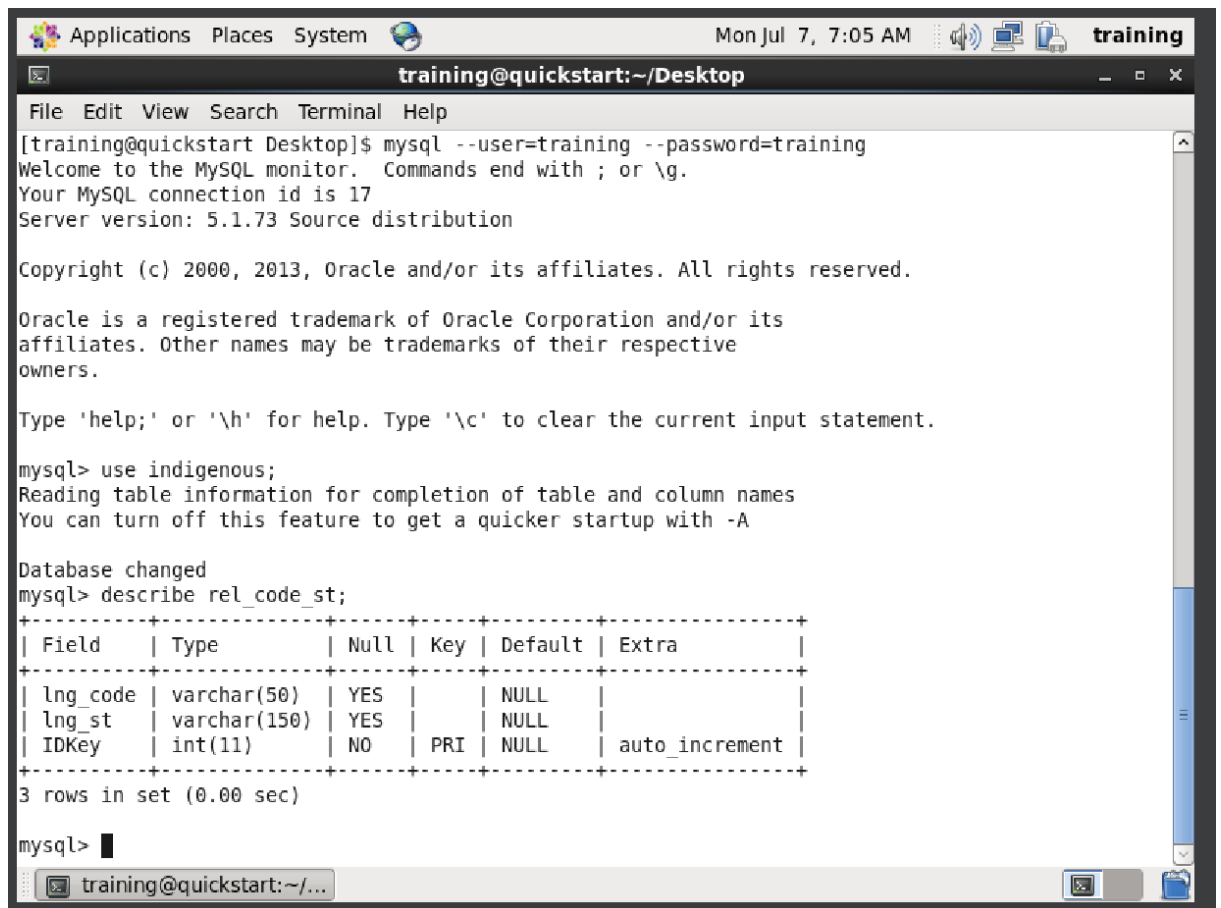
-- Create table rel_code_thl
CREATE TABLE rel_code_thl (
  lng_code STRING,
  lng_thl STRING,
  IDKey INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t'
LINES TERMINATED BY '\n'
STORED AS TEXTFILE;

-- Load data into rel_code_thl
LOAD DATA INPATH 'a2_db/rel_code_thl.txt' INTO TABLE rel_code_thl;

-- Create table rel_code_thp
CREATE TABLE rel_code_thp (
  lng_code STRING,
  lng_thp STRING,
  IDKey INT
)
```



6. Show the detailed information of table rel_code_st in MySQL.
 - 6.1. Statement:
mysql --user=training --password=training
use indigenous;
describe rel_code_st;



The screenshot shows a terminal window titled "training@quickstart:~/Desktop". The user has executed the command `mysql --user=training --password=training`. The MySQL prompt is `mysql>`. The user has entered `use indigenous;` and `describe rel_code_st;`. The output shows the table structure for `rel_code_st` with 3 rows in set (0.00 sec).

```
[training@quickstart Desktop]$ mysql --user=training --password=training
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 17
Server version: 5.1.73 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

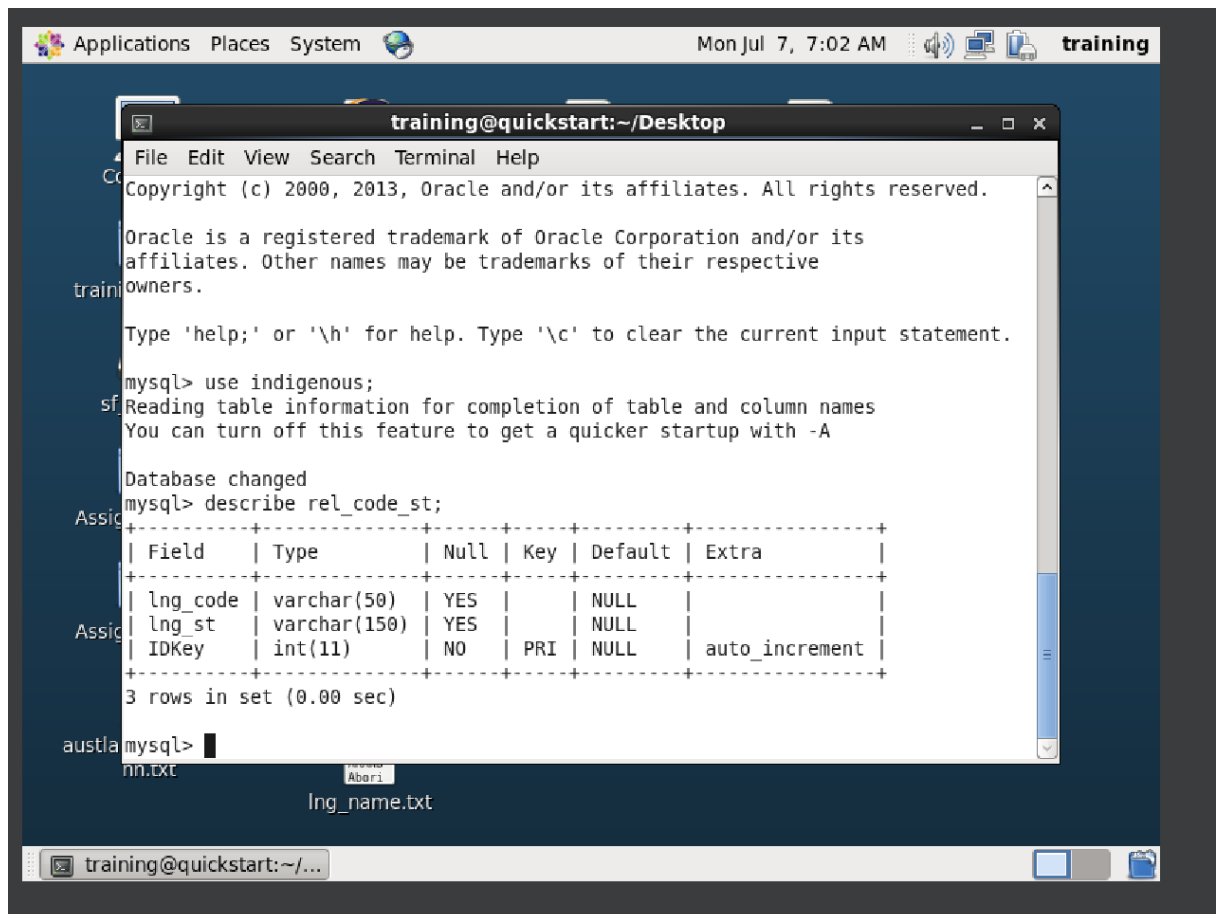
mysql> use indigenous;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> describe rel_code_st;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| lng_code | varchar(50)   | YES  |     | NULL    |                |
| lng_st   | varchar(150)  | YES  |     | NULL    |                |
| IDKey    | int(11)       | NO   | PRI | NULL    | auto_increment |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

Ans: _____

6.2. Result screenshot:



7. Collect the results containing "lng_code", "lng_name", and "lng_st" of tuples whose lng_name starts with the upper case "D". Show bot statements and the last page screenshot of your query result in MySQL and Hive.

7.1. MySQL query statement:

Ans:

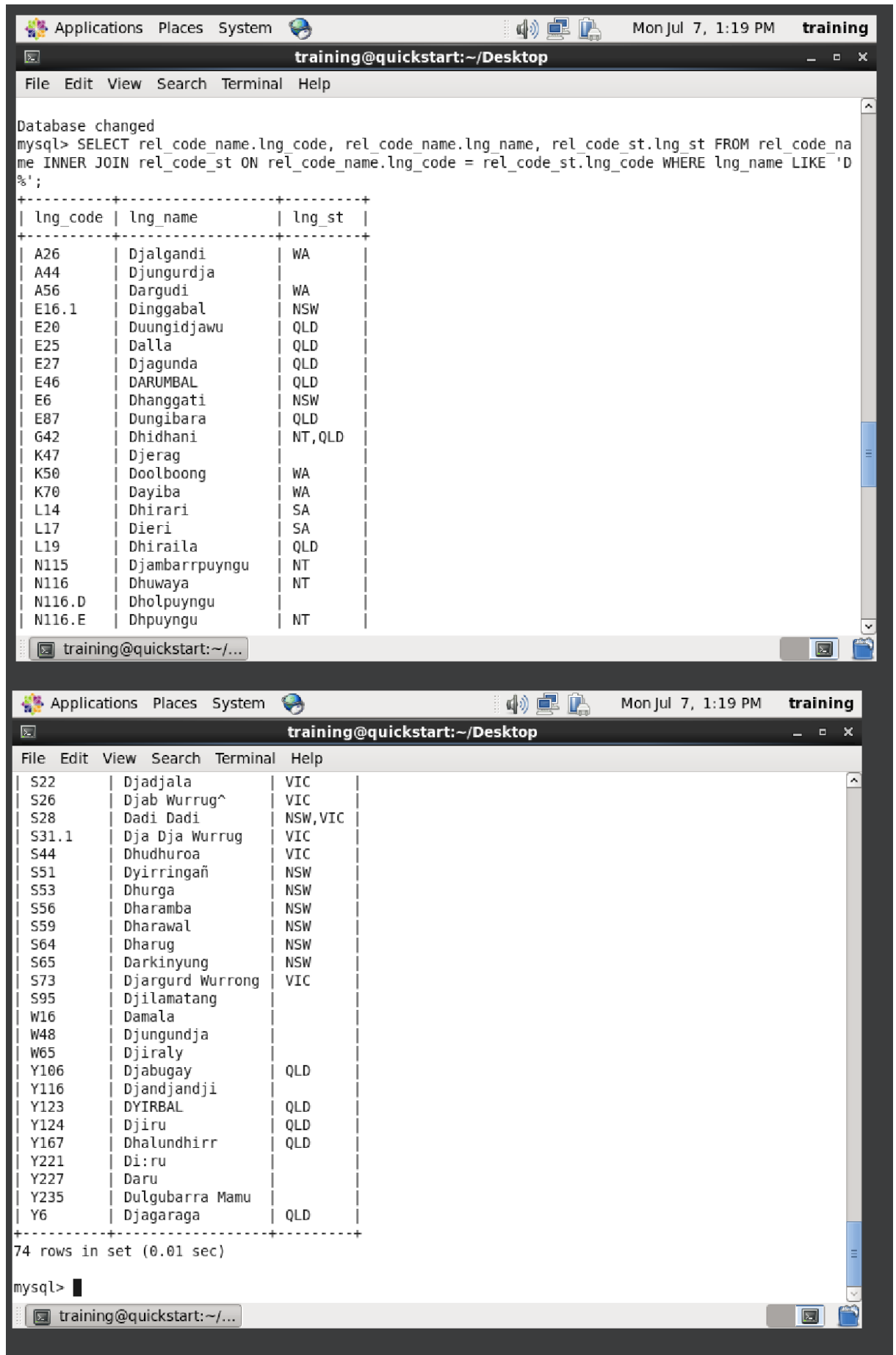
```
mysql --user=training --password=training
```

```
use indigenous;
```

```
SELECT rel_code_name.lng_code, rel_code_name.lng_name, rel_code_st.lng_st FROM
rel_code_name INNER JOIN rel_code_st ON rel_code_name.lng_code =
rel_code_st.lng_code WHERE lng_name LIKE 'D%';
```

7.2. MySQL result screenshot:

Ans:



The image shows two screenshots of a terminal window titled 'training@quickstart:~/Desktop'. The window displays a MySQL query and its results.

Top Screenshot:

```
Database changed
mysql> SELECT rel_code_name.lng_code, rel_code_name.lng_name, rel_code_st.lng_st FROM rel_code_name
INNER JOIN rel_code_st ON rel_code_name.lng_code = rel_code_st.lng_code WHERE lng_name LIKE 'D
%';
```

lng_code	lng_name	lng_st
A26	Djalgandi	WA
A44	Djungurdja	
A56	Dargudi	WA
E16.1	Dinggabal	NSW
E20	Duungidjau	QLD
E25	Dalla	QLD
E27	Djagunda	QLD
E46	DARUMBAL	QLD
E6	Dhanggati	NSW
E87	Dungibara	QLD
G42	Dhidhani	NT,QLD
K47	Djerag	
K50	Doolboong	WA
K70	Dayiba	WA
L14	Dhirari	SA
L17	Dieri	SA
L19	Dhiraila	QLD
N115	Djambarrpuyngu	NT
N116	Dhuwaya	NT
N116.D	Dholpuyngu	
N116.E	Dhpuyngu	NT

Bottom Screenshot:

```
mysql> SELECT rel_code_name.lng_code, rel_code_name.lng_name, rel_code_st.lng_st FROM rel_code_name
INNER JOIN rel_code_st ON rel_code_name.lng_code = rel_code_st.lng_code WHERE lng_name LIKE 'D
%';
```

lng_code	lng_name	lng_st
S22	Djadjala	VIC
S26	Djab Wurrug	VIC
S28	Dadi Dadi	NSW,VIC
S31.1	Dja Dja Wurrug	VIC
S44	Dhudhuroa	VIC
S51	Dyirringa	NSW
S53	Dhurga	NSW
S56	Dharamba	NSW
S59	Dharawal	NSW
S64	Dharug	NSW
S65	Darkinyung	NSW
S73	Djargurd Wurrong	VIC
S95	Djilamatang	
W16	Damala	
W48	Djungundja	
W65	Djiraly	
Y106	Djabugay	QLD
Y116	Djandjandji	
Y123	DYIRBAL	QLD
Y124	Djiru	QLD
Y167	Dhalundhirr	QLD
Y221	Di:ru	
Y227	Daru	
Y235	Dulgubarra Mamu	
Y6	Djagaraga	QLD

74 rows in set (0.01 sec)

```
mysql>
```

7.3. Hive query statement:

Ans:

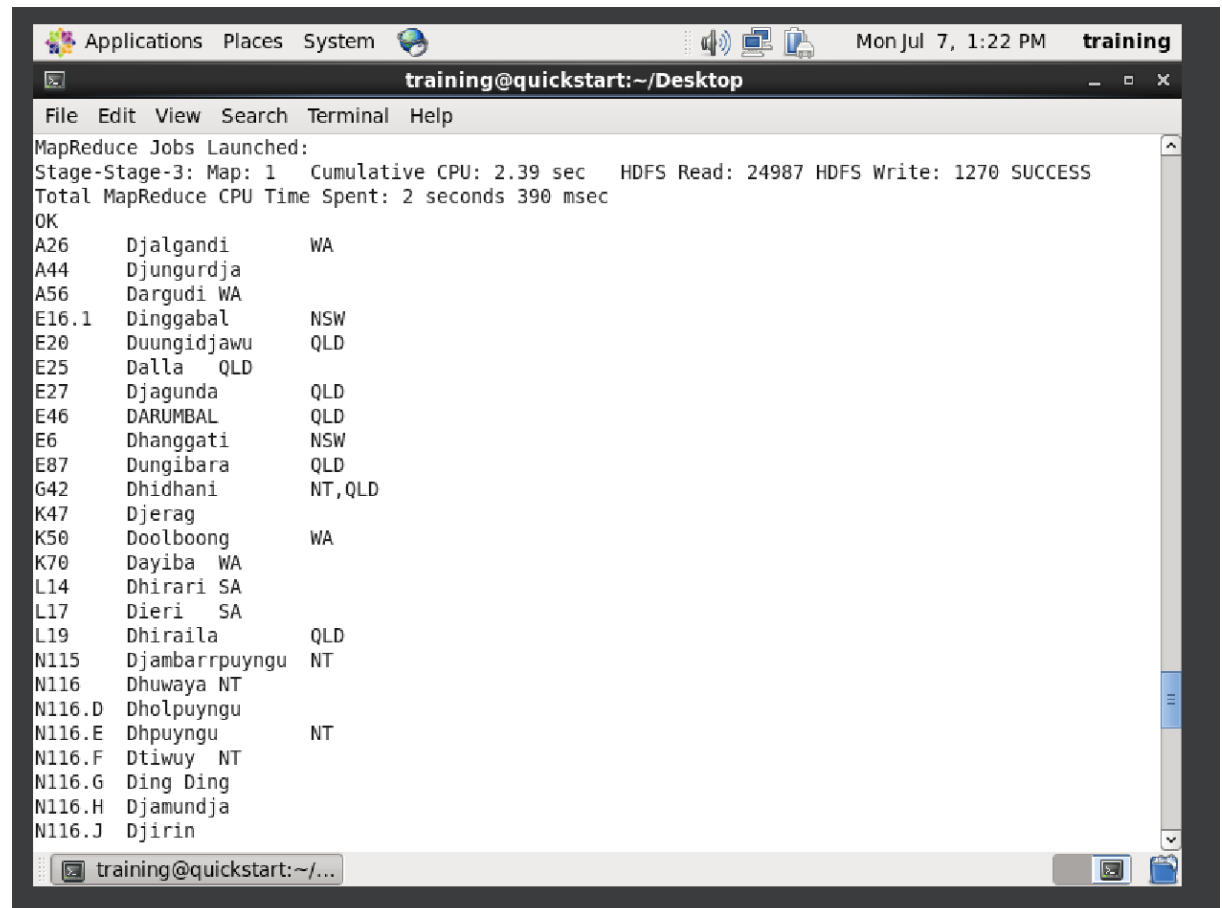
hive

use indigenous;

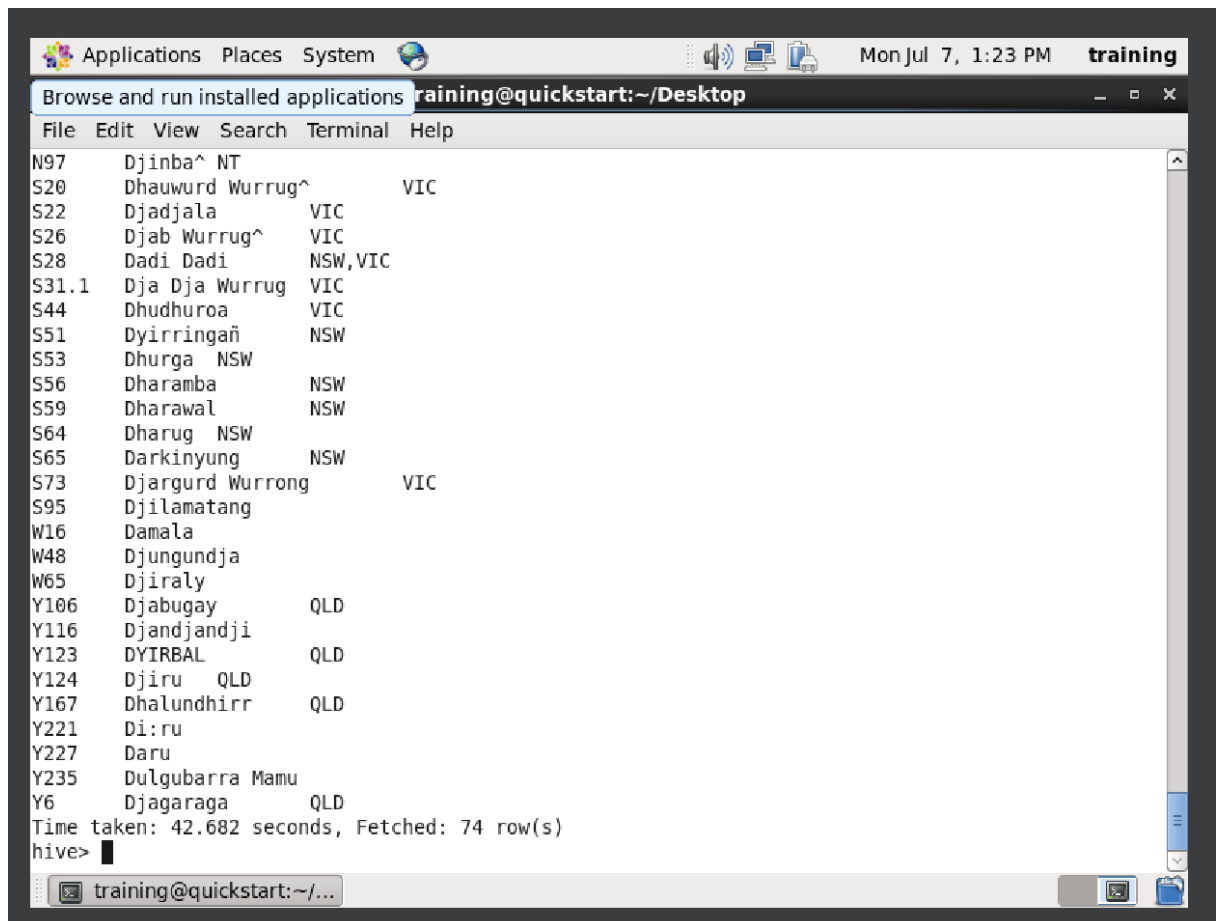
```
SELECT rel_code_name.lng_code, rel_code_name.lng_name, rel_code_st.lng_st FROM  
rel_code_name INNER JOIN rel_code_st ON rel_code_name.lng_code =  
rel_code_st.lng_code WHERE lng_name LIKE 'D%';
```

Hive result screenshot:

Ans:



```
training@quickstart:~/Desktop
File Edit View Search Terminal Help
MapReduce Jobs Launched:
Stage-Stage-3: Map: 1 Cumulative CPU: 2.39 sec HDFS Read: 24987 HDFS Write: 1270 SUCCESS
Total MapReduce CPU Time Spent: 2 seconds 390 msec
OK
A26 Djalgandi WA
A44 Djungurdja
A56 Dargudi WA
E16.1 Dinggabal NSW
E20 Duungidjawa QLD
E25 Dalla QLD
E27 Djagunda QLD
E46 DARUMBAL QLD
E6 Dhanggati NSW
E87 Dungibara QLD
G42 Dhidhani NT,QLD
K47 Djerag
K50 Doolboong WA
K70 Dayiba WA
L14 Dhirari SA
L17 Dieri SA
L19 Dhiraila QLD
N115 Djambarrpuyngu NT
N116 Dhuwaya NT
N116.D Dholpuyngu
N116.E Dhpuyngu NT
N116.F Dtiwuy NT
N116.G Ding Ding
N116.H Djamundja
N116.J Djirin
```



8. Collect the results containing “lng_code”, “lng_name”, “lng_st”, “a_lng_lat”, and “a_lng_lng” of tuples whose lng_synonym contains “Kerama”. Show both statements and the last page screenshot of your query result in MySQL and Hive.

8.1. How many tuples are retrieved at the end?

Ans: _____

9 tuples based on result

8.2. MySQL query statement:

Ans:

```
SELECT
lng_id.lng_code,rel_code_name.lng_name,rel_code_st.lng_st,lng_id.a_lng_lat,lng_id.a_lng_lng
FROM lng_id INNER JOIN rel_code_name ON lng_id.lng_code =
rel_code_name.lng_code INNER JOIN rel_code_st ON lng_id.lng_code =
rel_code_st.lng_code INNER JOIN rel_code_synonym ON lng_id.lng_code =
rel_code_synonym.lng_code WHERE lng_synonym LIKE '%Kerama%';
```

8.3. MySQL result screenshot:

Ans:


```

training@quickstart: ~/Desktop
File Edit View Search Terminal Help
_lng_lng FROM lng_id INNER JOIN rel_code_name ON lng_id.lng_code = rel_code_name.lng_code INNER J
JOIN rel_code_st ON lng_id.lng_code = rel_code_st.lng_code INNER JOIN rel_code_synonym ON lng_id.l
lng_code = rel_code_synonym.lng_code WHERE lng_synonym LIKE '%Kerama%';
ERROR 1046 (3D000): No database selected
mysql> use indigenous;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT lng_id.lng_code,rel_code_name.lng_name,rel_code_st.lng_st,lng_id.a_lng_lat,lng_id.a
_lng_lng FROM lng_id INNER JOIN rel_code_name ON lng_id.lng_code = rel_code_name.lng_code INNER J
JOIN rel_code_st ON lng_id.lng_code = rel_code_st.lng_code INNER JOIN rel_code_synonym ON lng_id.l
lng_code = rel_code_synonym.lng_code WHERE lng_synonym LIKE '%Kerama%';
+-----+-----+-----+-----+-----+
| lng_code | lng_name | lng_st | a_lng_lat | a_lng_lng |
+-----+-----+-----+-----+-----+
| W31      | Yarnarri | WA     | -22.3454093 | 117.1252561 |
| W36      | Kurrama  | WA     | -21.84552987 | 116.1884783 |
| W49      | Jadira   | WA     | -17.36377694 | 145.7047546 |
| Y121     | Ngadjon  | QLD    | -17.63998402 | 145.7485185 |
| Y122     | Mamu     | QLD    | -17.74022656 | 145.661334 |
| Y123     | DYIRBAL  | QLD    | -17.74022656 | 145.661334 |
| Y123     | JIRRBAL  | QLD    | -17.90722366 | 146.0521375 |
| Y124     | Djiru    | QLD    | -17.96177304 | 145.8525333 |
| Y126     | Gulngay  | QLD    |              |              |
+-----+-----+-----+-----+-----+
9 rows in set (0.43 sec)

mysql>

```

8.4. Hive query statement:

Ans: SELECT

```

lng_id.lng_code,rel_code_name.lng_name,rel_code_st.lng_st,lng_id.a_lng_lat,lng_id.a_lng
_lng FROM lng_id INNER JOIN rel_code_name ON lng_id.lng_code =
rel_code_name.lng_code INNER JOIN rel_code_st ON lng_id.lng_code =
rel_code_st.lng_code INNER JOIN rel_code_synonym ON lng_id.lng_code =
rel_code_synonym.lng_code WHERE lng_synonym LIKE '%Kerama%';

```

8.5. Hive result screenshot:

Ans:

```

le02--.hashtable (31237 bytes)
2025-07-07 01:31:32      End of local task; Time Taken: 3.537 sec.
Execution completed successfully
MapredLocal task succeeded
Launching Job 1 out of 1
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1751908907534_0009, Tracking URL = http://quickstart.cloudera:8088/proxy/appli
cation_1751908907534_0009/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1751908907534_0009
Hadoop job information for Stage-5: number of mappers: 1; number of reducers: 0
2025-07-07 13:31:47,882 Stage-5 map = 0%,  reduce = 0%
2025-07-07 13:32:01,674 Stage-5 map = 100%,  reduce = 0%, Cumulative CPU 2.88 sec
MapReduce Total cumulative CPU time: 2 seconds 880 msec
Ended Job = job_1751908907534_0009
MapReduce Jobs Launched:
Stage-Stage-5: Map: 1  Cumulative CPU: 2.88 sec  HDFS Read: 222993 HDFS Write: 341 SUCCESS
Total MapReduce CPU Time Spent: 2 seconds 880 msec
OK
W31      Yarnarri
W36      Kurrama WA      -22.3454093      117.1252561
W49      Jadira WA      -21.84552987      116.1884783
Y121     Ngadjon QLD      -17.36377694      145.7047546
Y122     Mamu QLD      -17.63998402      145.7485185
Y123     DYIRBAL QLD      -17.74022656      145.661334
Y123     JIRRBAL QLD      -17.74022656      145.661334
Y124     Djiru QLD      -17.90722366      146.0521375
Y126     Gulngay QLD      -17.96177304      145.8525333
Time taken: 42.931 seconds, Fetched: 9 row(s)
hive>

```